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Lecturer: AMI Elementary Trainer, Phyllis Pottish-Lewis

Introduction to Biology

As you unfold the universe to the child, a part of that has to be biology. And of course biology consists of plants and animals - those creatures that you have heard called the furnishings of the earth. Life on earth is a natural part of our environment. It is part of the child's existence. And that is why we introduce our children to the world of biology.

We treat biology consistent with the principles of cosmic education in that it is part of the vision that we want to unfold to the child. It is not treated as a subject unto itself but it is treated as part and parcel of the process of life on earth. So, when we introduce the plants and the animals, we introduce them in relation to those factors that influence them and those on which they have an influence. In this way, we put biology in its natural perspective. This perspective **shows that there is a mutual relationship between the earth and all of the life on earth.**

As we unfold this chapter to the children, it becomes obvious to them that other chapters must have preceded life on earth because the earth needed to be prepared in order to support life. There is at least one other reason to present plants and animals to children as well. It provides them the opportunity to express their human tendencies.

Biology in the Casa dei Bambini

In the Casa, the child has had many opportunities with plants and animals to meet the human tendency to explore. His exploration was a sensorial exploration. As the children have been exploring in the primary class, they've had many opportunities to observe carefully both plants and animals. Not only has the child had the opportunity to explore and observe plants and animals in the Casa, but also at home. And these plants and animals, for this first plane child, have existed as sound and color or different named and unnamed, sometimes unorganized qualities in the environment. When the child comes to the Casa, he is given opportunities to clarify these impressions.

One of the ways is to give the names or the vocabulary that goes with these impressions. The vocabulary becomes a tool by which he can order his mind and thus build his intellect. Each new word allows the child to expand on his impressions. Once he builds up impressions, he is able to classify them and it is through this process of classifying that the child orders his mind, and by ordering his mind, he is building his intellect and becoming acquainted with natural patterns observed in the world. This

process of classification also satisfies the human tendency to order.

When there are plants and animals in the primary classroom, the children will derive all kinds of impressions related to them. The child will have seen different kinds of animals – ones that have backbones, ones that don't have backbones, animals that live in water and animals that don't live in water. In the area of practical life, the children will have had experience caring for the plants. Perhaps they will have grown seeds, thereby experiencing the complete life-cycle of a plant. Perhaps they use them in their artwork, observe them to draw them. And of course, they may have had experience caring for both the plants and animals, feeding and watering the animals, cleaning their cages, watering the plants and washing their leaves.

From all of these experiences and firsthand activities, the children will be aware that both plants and animals have needs and that in order to survive, these needs must be met. Animals must have food and water. Plants must have water and sunshine. The children become aware of these needs, not because they have been lectured to about them, but because they have experienced them directly.

Since the child in the first plane is a sensorial explorer, we only need to give one key to encourage that exploration. Once that key is given, the child will naturally explore.

The **biology cabinet is one such key**. It is the key to the different shapes of the leaf. Once the child works with the botany cabinet in the classroom, he has a key to help find those same shapes out in nature. As he is observing nature, he will discover that there are shapes that are not found in the botany cabinet. He will find that an interesting fact. He has made a discovery. They usually think you know nothing about it.

As the children are observing all of the shapes, very gradually, they become aware that leaves have different colors too. This is exciting for them. They continue to explore and their eyes might fasten on the flowers of the plant. And because the child is attuned to shape, he will begin to notice that **corollas** have different shapes. And they have different colors too. And so his exploration continues but it has been triggered by one key – the **botany cabinet**.

This same awareness happens in regard to animals as well. The children discover that animals have different shapes. They begin to notice the different skin colorings. They have different colors, different textures, and animals make different sounds. All of this awareness come to the child from his own exploration and from his own experience. It is all of these experiences that build a storehouse of information that the child will take with him to the elementary class.

Once the child gathers from his sensorial exploration this fund of knowledge, he can order and classify it. In the case of the leaves, we aid this process by **giving the name of the leaf shape, rather than giving the name of the plant** from which the leaf comes. The name of the shape gives him information that can eventually

be ordered, for example, this plant and this tree both have ovate leaves – but these plants over here have elliptical shaped leaves. The names of the plants alone give no means for ordering and classifying.

The fund of information is also built through the **nomenclature material**. The child learns the names of plants and animals, also the parts of plants and animals – and eventually this information can be ordered and classified. In the primary or prep course, you heard of language training and enrichment of vocabulary. Both of these areas contribute to the fund of information that the child is building. He can learn facts and information about plants and animals just as easily through stories, songs and poems, as he can from the materials.

Biology at the Elementary

At the second plane, we continue to look at the human tendencies that are continuing to operate but we have to look at the changing psychological characteristics of the elementary child. We then need to adapt our treatment of plants and animals to the newly emerging characteristics. The child continues to explore but now he is exploring with his **imagination**. Now the child has a **reasoning** mind. So, he uses his reason to assist him in his explorations. Now he is interested in the 'how' and 'why' of things. He is not fascinated merely by facts and names any longer. He is not interested in the fact that the leaf is green. He wants to know why the leaf is green. The fish is dead; why is the fish dead? He wants to know the reasons behind these facts.

Because the elementary child is now using his reason, when we introduce the plants and animals to the children, **we must always focus on the function of what the part does**. Or, in other words, we have to give the **physiology** of the part in focus. This is critical because it is the function that explains the reason for the behavior. For example, the **leaves are the part of the plant that make the food** for the whole plant. Since they need light to perform this process, **they grow towards the light**. The **roots have the job of providing water** that is needed to make the food and so the roots **grow towards water** so they can provide it for the leaves. So, from the children's work with plants and animals, it becomes apparent to them that the behavior of the plant or animal is directly related to the function that it must perform in order to meet the needs for survival.

Since the child is now exploring with his imagination, **allegory and imagery** appeal to this child, especially the 6 and 7 year olds. The children are delighted by stories about plants and animals and because of this, the teacher becomes a story-teller of the truth. Initially, the stories are simple – with the whole being given first and details added later in follow-up stories. To assist the imagination as we tell stories, we also introduce impressionistic charts or we do experiments and demonstrations that are related to our stories. When the children comprehend the concepts that we introduce, through their imagination, it begins to arouse an appreciation in them for the way plants and animals work. Imagine all of the different ways that plants grow towards the light. It is

fascinating. The children come to see how clever the natural world is. And by seeing this, they begin to appreciate the world of nature.

Another thing the children come to appreciate by observing nature is that there is a great **diversity**. All plants need light but they all don't get it in the same manner. The diversities become a point of interest for the children. Once you give the children the key, for example, that plants need sunlight to make their food, you can send them out to explore the varied and clever ways that plants have devised to meet this need. So, from their own explorations, motivated by their own interest, the elementary children observe the different solutions that plants have derived in response to the same need. The awareness of diversities now allows the child to classify and order his new information. So, here is another way the plants and animals help the child to meet the tendency to order. All the animals that protect their bodies with hair belong in this group, while all that protect their bodies with feathers, belong in another groups.

This **ordering and classifying** of characteristics leads to another kind of exploration. This new kind permits the child to **analyze** a specimen based on its characteristics, thereby developing his analytical abilities. It also allows him to categorize according to scientific classification. For example, this animal has feathers and so we put it into a group that we call birds. Or, this plant has parallel veins, so we put it in a group called Monocotyledons.

This activity of classifying allows for further development of human potential. The child is making choices by analyzing, by reasoning things out and he is classifying his impressions into an orderly structure – a process which corresponds to the order of his mind. And so through this whole process, in connection with plants and animals, he is building his intellect and further developing his human potential – for reasoning, analyzing and making critical choices.

The structure of biology in the elementary class is created from **observing the needs of plants and animals**. What do they eat and how often? How often do they need water? How do they move? It is from their observations that they understand the physiology. The **plant begins to droop because it needs water**. The reptile becomes sluggish when his environment is too cold. Since the structure of biology is based on observation, it is crucial that your children have the opportunity to observe. From these opportunities to observe, they will **develop and refine their powers of observation**. And this skill of observation will be used in all areas of their lives – not just with plants and animals. Observation also helps to sustain the interest that has been aroused from your stories and you want to make sure that you keep the interest and the excitement alive in your children. *Otherwise, they get bored and when they get bored, they can get into trouble*. So, you want to keep the children excited through their observations and so you have to ensure that they have a means to go out and observe the world of nature. It is only in nature that the child will make further discoveries. It also helps the child to meet his tendency to explore. The going out is critical for the following reasons:

- He needs to have an **opportunity to follow interest**
- He needs to become **aware of the immense diversity in nature**

- He needs to **personally experience how clever plants and animals are** in devising ways to meet their needs

This will create an appreciation for life on earth. As the children explore themselves in nature, they soon discover there is a delicate balance on earth between the earth itself and the furnishings of the earth. We hope the child realizes that because the balance is so delicate, humans must work to preserve it – and they must act responsibly in relation to nature. This understanding is more important than learning the shapes of the leaves or the names of the plants.

We help the children become responsible to nature by helping to **develop their ability to be responsible**. In biology, they can assume many tasks in relation to the plants and animals in order to develop responsible behavior. They should assume the care for all the animals and plants in the classroom and by so doing, they develop a degree of responsibility that one day they can apply to the environment at large.

With guidance, the child begins to see the reasons behind the things that he has observed. He becomes aware that plants and animals take from the earth but also give back. This **process of giving and taking** that each organism performs is like a **cosmic task**. Our hope is that from this awareness, the child will realize that he too has a task, a contribution to make, and so begin to make meaningful choices to fulfill that task – in whatever way it is envisaged by the child.

Since the child's sense of responsibility has been built by degrees, when it comes time for him to act responsibly in nature, he will be prepared with the knowledge and understanding of all the factors involved. He will know that this is a world that has taken many millennia for life to adjust, for a balance to be reached. He will also know that humans, unlike the animals, have the power to alter the nature of their world. He will also know that life and nature are continually asked to make adjustments - not merely to the copper or the rest of the minerals that are washed from the rocks and carried to the sea - but adjustments to the synthetic creations of humans inventive minds as well. The awareness of all of these factors will make the child realize that if he must disturb these delicate relationships, he must do so thoughtfully and responsibly, with full awareness that what he does may have consequences remote in time and place.

October 18, 2004

The Needs of the Plant

Specific Needs

Setups

- Four containers of equal size
- Cotton balls
- Fast-growing seeds (radish, alfalfa, cress, mustard) – make sure seeds are viable!
- Tickets
- pencil
- water
- hand-towel



Prologue

Though we could take animals first or present animals and plants in parallel, we begin here with plants. They are easily observed and this is an obvious advantage.

This will be presented in the first weeks of a 6 to 9 year old class. It is possible to be presented in the primary however. It is a critical activity because it is a key to understanding many things about the different parts of the plant. Don't under-estimate its value.

We gather a group of newly arrived children. Have them help you collect the materials. Let them do the activity so it is theirs and not yours. Once we see results after 1-3 days, gather the same children and with them analyze what happened. Plant around 10 seeds per container.

Do not tell them the conclusion ahead of time!

Begin

First thing to do is to put some cotton balls in the containers. Make sure they have a little bed of cotton and then place the seeds in the cotton bed.

Water all containers except one. **This one gets no water.**
(let the children make a ticket saying 'no water')

And let's put this one in the sun. (place first container on the window ledge)

This one, we're going to **deprive of light.**
(let the children make a ticket saying, 'no light', for the second container)

Where can we put this so it will get no light? I know - let's place it in the filing cabinet!
(you can also put a container over the top to block all light)

This one, we'll **deprive it of warmth.** (children make ticket saying 'no warmth', for third)
I know where I'm putting this one – outside. We'll need to check this one later outside.

This last one **gets everything** it needs. (children make a ticket saying 'gets everything')
It will get warmth, sun and water. (place fourth and last container on window sill)

Now let's watch and see what happens. We have to keep our eye on this. Of course it won't happen today, but **each day we should check the plants** to see what happened since we last looked.

(at end, after results are reached)

For a plant to be healthy it needs water, warmth and light.

Epilogue

The seeds that got no water did not even germinate. The seeds that were put into the cold did germinate but they were stunted in their growth. The seeds that got no light grew quite tall but their leaves are yellowish – not green. Why did they grow tall? (They looked for the light.) The one that got everything – it is healthy.

October 18, 2004

Plants Grow to the Light

Setups

- Square container
- Pot shards (or rocks or charcoal from an aquarium – i.e., drainage materials)
- Soil
- Fast-growing seeds (radish)
- Wooden box that corresponds in size to the square container with two windows

Prologue

Put drainage material in the square container. Cover it with soil. Over the top of the soil spread many radish seeds. Water it, place it in a warm place with light so it will grow up. We want to have a little forest of radish seeds.

Once they are grown up (about 2 inches), gather children who've had the first activity, so they now can again see that plants need light.

Illustration

Place box over the square container. Rim of the square container corresponds to the bottom of the rim of the windows. (Rim of container corresponds in height to the bottom rim of the window.)

Begin

(show them the forest of plants, place it in the light with one window open and the other closed)

(Invite the children to wait and watch. On a sunny day it will take just a few hours to get results.)

Keep your eye on this all morning. Check on these plants before you go out to play at lunchtime. I think something will happen.

(when children check on them, they will see the little plants incline themselves out the window)

It seems like the plants grow toward light! Let's see about this.

(close up the open window and open the other window)

Let's wait a while to see what happens.

(the children will see the **plants make a u-turn and grow in the opposite direction.**)

Why did this happen? Because plants grow toward light.

Epilogue

This is another activity that should be presented in the first weeks of school. It is a basic awareness they need to have.

October 18, 2004

Plants Need Minerals

Setup

- About 10 pea seedlings that have been germinated in sawdust
- six tall containers, the taller the better (such as a tall peanut butter jar)
- Six disks attached to the top of the container. (in center is a little hole with a cut from the circumference to the center)
- Black paper corresponding in size to the area of your jar side so that it blocks light
- Six rubber bands to attach the black paper of the jar
- “Needs of the Plant” chart A

Sawdust is a good material to use because it is clean and is easy to extract roots out of the medium.

Make 6 solutions of distilled water and:

- 1 gram of calcium nitrate to one liter of water
- ½ gram of potassium phosphate
- ½ grams of magnesium sulfate (also known as Epsom salts)
- A few drops of ferric chloride



Prologue

This lesson comes at the end of the second year or beginning of the third year (8 to 9 years old).

These solutions can be made with the children or beforehand:

One container will have all minerals, in amounts listed, in one liter of water.

One container will have none of these minerals.
One container will have all but calcium-nitrate
One container will have all but potassium-phosphate
One container will have all but magnesium-sulfate
The last container will have all but ferric chloride

Material Making

Illustrate the “Needs of the Plant” chart and color-in the actual chart.

Begin

(make up the solution)

Put water in the container. Take seedling, place it in the disk so root is below and other is atop.

Place it in the water and attach it. (Jar top holds seedling up with root hanging below)

Put black paper around it and attach rubber band to hold it in place against outside of jar.

Write a ticket saying 'no minerals' for the jar with that solution. Place them together.
With solution that has all minerals, in manner above, make ticket that says 'all minerals.'
With third solution, do same as above with ticket 'lacks calcium nitrate.'
With third solution, do same as above with ticket 'lacks magnesium sulfate.'
With third solution, do same as above with ticket 'lacks potassium phosphate.'
With third solution, do same as above with ticket 'lacks ferric chloride.'

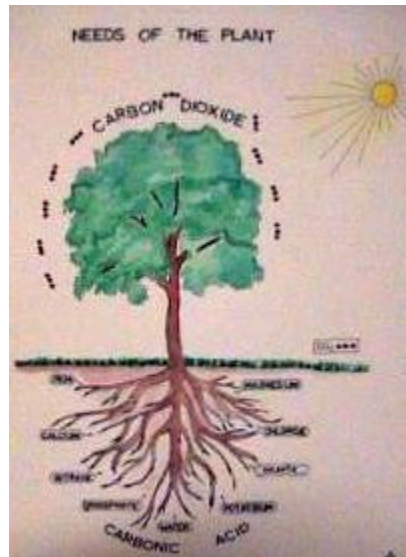
Set them in the sun, watch and wait. This time, we have to **watch for about 3-4 weeks**.

Results Summary

Gather the children and observe the growth in each jar. Analyze what has happened.

The container that had no minerals did not grow well compared to the rest.
The one that had all the minerals grew the best.
The one that had ferric chloride has whitish leaves, less green.
All the others will be plants that look unhealthy compared to the one with all the minerals.

(show them “Needs of the Plant” chart (chart a) after doing this summary)



For a plant to grow up and be healthy, it requires sun, (point to chart), water (point), carbon-dioxide (point to atmosphere) and it also requires iron, calcium, nitrate, phosphate potassium, sulfate, chloride and magnesium (point to roots).

It takes all of these things to make a healthy plant.

Epilogue

Make sure to put more solution as the liquid evaporates!

Can you engage all and see their eyes? That is the key to knowing how many children to include.

October 18, 2004

The Leaf

The Function of the Leaf

Material

chart B and chart C

Prologue

We tell stories, not merely inform. We begin with the main function and since we also appeal to the reason, we begin with a question whenever possible.

Begin

How do plants eat? Each leaf contains water. Each drop of water consists of two particles of hydrogen and one particle of oxygen. With the help of the sun, the green color in the leaf separates the oxygen from the hydrogen. The hydrogen is then taken and mixed with the gas that has come in from the air called carbon dioxide. Then these two are cooked into plant food. The oxygen that is not used is thrown out of the leaf as a waste. And it is in this way that the leaf makes the food for the whole plant. So, the leaf is like a food factory.

(Now that the story is told – not before, introduce chart b – point to each step in the chart as you retell the same story.)



It's like you see here. Water is brought up from the ground and each particle of water is made up of two hydrogen particles and one oxygen. With the help of the sun the green stuff in the leaf separates the hydrogen from the oxygen.

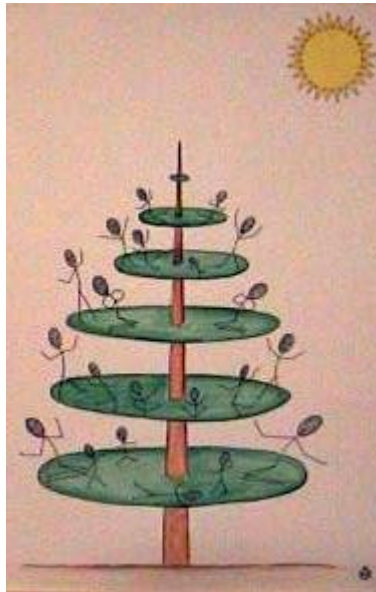
The hydrogen particles are then mixed with that gas that comes in from the air, and

together they are cooked into plant food. The oxygen that is not used is thrown off as a waste.

And that is why the leaves will grow towards the light – because they need the light to help them make their food. If you notice carefully the plants outside, you'll notice they arrange themselves to get the most light possible.

Would you like to go outside and look at some plants? Okay, see you.

(Once they come back into the classroom, and they're talking about their experiences, introduce them to chart c)



Look at this. Each plant is like a temple to the sun. And each leaf is like a priestess worshipping the sun – because it is the sun on which they depend for life.

Sometimes, plants are even shaped like a temple (hold up chart)

Epilogue

The story was simple. You tell it before showing the chart. You appeal to imagination and reason. And we brought out several important points: Why leaves grow towards light and that the leaf is the food factory.

Notice, we said nothing about chlorophyll nor photosynthesis. We'll take up these terms later. Leave the chart out so they can look at it, talk about it and think about it. Notice we send them off on their own – looking for the ways that plants grow toward the sun. You want them to make certain discoveries and so you can guide their

explorations.

On some plants, like **hedges**, leaves are placed only on the outside where they receive sun. There are no leaves in the inside of a hedge because no light goes into the hedge. Examples of this are beeches and holly.

Another element that can be pointed out is that some tall plants have their **longer branches on the bottom** and shorter branches on the top, so the light can reach all of the branches. Examples are pine and fir trees.

Another aspect: sometimes in smaller plants, the **leaves are arranged very carefully so that each one can get some light**. Examples are geraniums and ivy.

Also, we should point out they **struggle towards the light**. Some little plants just don't get enough and so they look miserable and unhealthy.

Another aspect to highlight is that some plants are able to grow to the sun because they have devised **ways to climb**. They usually climb on taller plants or even on buildings. These are the plants that **twine around and hook** onto things.

Also, point out that some plants that don't have tidy arrangements and can't compete well for light, **come out very early in the year before the giant trees** take up all of the sunning area. The plants that do this are primrose – which means first rose, or milkmaid – because she gets up early in the morning to do her work, or snowdrops – that come out so early there is still snow on the ground, and finally, dogwood.

You don't have to point out all of these aspects immediately but these are aspects you ultimately want them to become aware of. They just need to have some exploration done so that they're consciously **aware that plants grow toward the sun**.

Chart C elicits much conversation. Ask them, “what did you see”? You need to know your environment a bit.

October 18, 2004

The Stomata

Stomata's Main Function

Material

chart D, a well-water plant, a plastic bag and a rubber band.

Prologue

How does that carbon dioxide get into the leaf?

Begin

On the underside of every leaf, there are tiny little pores and they have the shape of a mouth.

They are given the name stomata. And like so many mouths, they open and they close. (open and close your mouth – they love it)

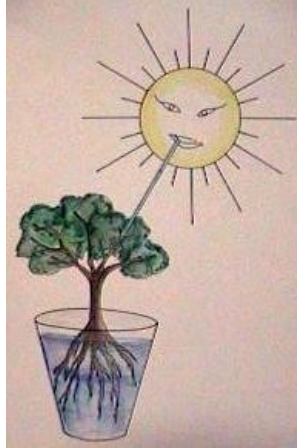
The mouths are either open or closed depending on whether the leaf needs water or not.

**If the leaf wants to save water, the stomata are closed.
And if the leaf wants to get rid of water, the stomata are open.**

This is how the plants evaporate. (Botanists say the plant evaporates though it is actually the water)

Do you remember how hot and humid it was when we visited the hot-house. That is because the leaves are giving off their water.

(present chart D)



As the water goes out of the leaves, it seems as if the sun has a good drink. The sun drinks from all the plants of the world. (set chart on display to the side)

This is why the forest of woods is full of moisture. And that is why woods are said to attract rain. There is a lot of moisture in the air. In places like deserts, where there are fewer plants, the area is dry because there are not a lot of plants to give off water so rain can be formed.

Let's see what happens when we place the plastic bag over the plant and secure it with the rubber band. We'll place it in the sun for a time. Let's watch and wait to see what happens.

(if the plant has enough water in it and it is a sunny day, it will take only a few hours before the plant gives off its water – evaporates – and water condenses on the inside of the bag. After you see condensation, gather the children to the plant, carefully take the bag off the plant, turn the bag over and flick it so all the water collects in one corner. This give a great impression – about a tablespoon of water sometimes is evident.)

Epilogue

The idea of **rain being precipitated by plants** is an important point to bring out in the theory of cosmic relation. The woods versus desert – there is a connection between plants and the environment.



Stomata Follow-Up with Microscope

Material

A broad-leaf leaf (geranium or onion leaves), a microscope slide, a cover slip, an eye dropper and some technique (pull at a diagonal).

Prologue

Find an area between the leaf. You want to tear it diagonally to see the epidermis. It is a very thin white layer within the leaf. We want the children to see the stomata. Take it and place the leaf part on the slide with some water on the slide also. Secure it with your cover slip and place it on the microscope. With luck you will see the stomata.

If you don't have a microscope, you can point them out in books.

October 18, 2004

Leaves Give Off Oxygen

Material

Large transparent container (e.g., a peanut butter jar), a transparent funnel, a test tube, and an aquatic plant (one that grows in the water such as Elodea or Anacharais)



Prologue

Fill the jar with water nearly to the top. Take the aquatic plant and place it in the mouth of the funnel. Place it all in the large jar with water – keeping the plant contained in the mouth of the funnel. (Stem of the funnel is under the water).

Take the test tube and fill it to the brim with water. Put thumb over the top and then put it under the water. Turn it upside down making sure no air enters the test tube. Then put the test tube over the stem of the funnel in the water.

Place this in the sun and watch and wait. After a few hours, the little plant will give off oxygen and the oxygen accumulates at the top of the test tube.

Epilogue

Let them verify there are no air bubbles at the first. This provides a dramatic impression once the air bubbles are detected. All of these presentations up to here have given the function of the leaf.

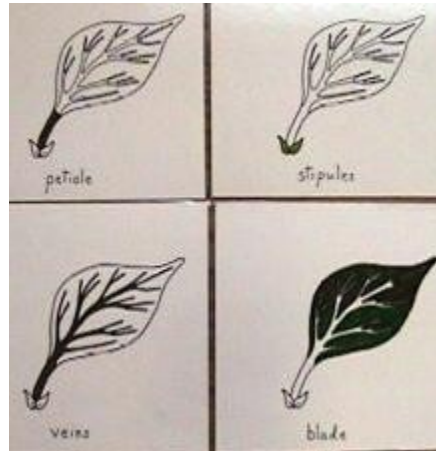
October 18, 2004

The Parts of the Leaf

Material

A leaf.

Petiole



Stipules

Veins

Blade

Prologue

When you present the parts of the leaf, pick a specimen that is large and not multi-colored. It should have no diversion to distract the child from the key points of the lesson. You could have the leaf in the classroom or do this outside.

Begin

Leaves, like the rest of the plant body, are made up of little things that are like the bricks of a building. These are called the cells. It is in the cells that the food is made. And the cells have to stay extended so they can get light from the sun.

The support to extend the cells is given by the **veins**. (point out the veins)

The veins crisscross through the whole leaf. They are made of a harder material than the other parts of the leaf. And they are hollow inside like pipes. **Like pipes, they carry water.**

The water is brought up from the ground and taken to the cells so that the leaf can make food. Once the food is prepared, it is carried down the veins to feed the other parts of the plant.

This soft part is called the **blade** (or the lamina, it has two names). These are the veins that expand the blade.

This part here (point to it) is called the **petiole**. The petiole helps to turn the blade to the sun. At the base of the petiole, we have small leaf-like parts called the stipules.

Epilogue

If you have a specimen with soft parts eaten away on the blade show them how the vein system expands the blade. Otherwise show them a picture in a book.

October 18, 2004

Varieties of the Leaf

Materials

See underlined.

Prologue

When you show varieties, give them a real specimen – not something from a book. The headings such as parallel veins are used as scientific classification. Once you have given the children the parts of the leaf, send them out to explore the varieties. You can begin the lesson in the classroom or if you are outside, begin it there.

Begin

Parallel and Reticulate Veins

Let's look at the veins on this plant. Do you notice that the veins go out in straight parallel lines?



Leaves that have veins like this are called parallel veins. (holds up a reed-like leaf)

How about this? This leaf has veins that go out and look like a net all through the leaf. This kind of vein system, we call reticulate.

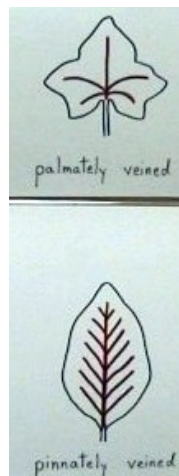


Reticulate comes from the Latin word 'rete' meaning net. So here we have a reticulate vein (hold up example) and here we have a parallel vein (hold up example)

Palmately Reticulate or Pinnately Reticulate

Gather the children with their collections.

These leaves have reticulate veins but they're a little different from each other.



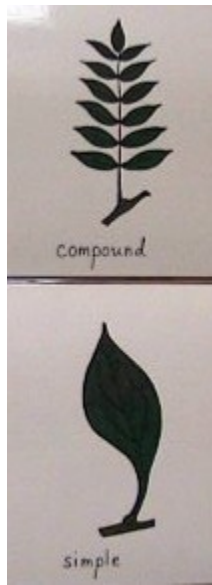
This one has the veins going out like a hand. It is called palmately reticulate – from the Latin word palma – meaning hand. (veins all come out from a single source point)

On this leaf, the veins go out like a feather so it is called pinnately reticulate – from the Latin word pinna – meaning feather.

Simple and Compound

(use samples from the children's collections)

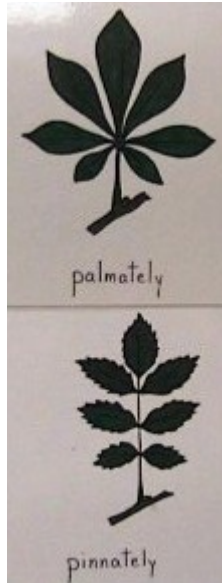
Let's look at these two leaves. This leaf does its work with leaflets (it has a system of 4 leaves) This is called a compound leaf.



This leaf doesn't have leaflets. This is called a simple leaf. (hold up leaf)

Pinnately Compound and Palmately Compound

Look at these two leaves. They are both compound. But they look a little different from each other, don't they? This one, where the leaves go out like a palm, is called palmately compound. (leaves go out in a circle almost)

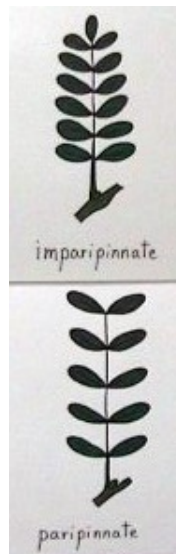


This one where the leaves go out like a feather is called pinnately compound.

Paripinnately Compound and Imparipinnately Compound

Look at these two compound leaves. They are pinnately compound aren't they?

This one has a pair of leaflets on the top. (hold up). It is called paripinnately compound.



This one has a single leaf at the top. It does not have a pair of leaflets. It is called imparipinnately compound.

Epilogue

Examples of parallel veins are lilies, spider plants, iris, plantain. Reticulate veins examples are dogwood, cottonwood, apples, strawberries and most everything actually. With these two types, we've given the children two very basic aspects. Send them out to make a collection. They will come back with a collection of leaves. You can use them to go on to the next characteristic – palmately reticulate or pinnately reticulate

Ivy, maples and geranium are examples of palmately reticulate. Examples of pinnately reticulate are oaks, elms, holly and birch.

Examples of compound leaves are roses, acacia, locust, schefelara, lupins and beans. Examples of simple leaves are oak, willow, peach, apple and maple.

Examples of pinnately compound are rose, ash, hickory and locust. Those that are palmately compound are chestnuts, lupins and schefelara (this is a good plant to have in the classroom because it is hardy).

Examples of imparipinnately compound are rose, locust and ash. Examples of paripinnately compound is Kentucky coffee and Ailanthus.

One hint about looking for **compound leaves**: some look like compound but they are not. The **leaflets have to be directly opposite each other** and another aspect that usually is the case, though not always, is that there is no petiole. The leaflets are attached directly to the stem. We have to train our eye to know for sure if they are compound.

These reticular characteristics are the characteristics they will use for classification and with classification guides.

Remember that when you introduce characteristics, introduce each in isolation versus how we've seen them today. When you send the children out to explore, you are guiding their observations. As one observes, gradually the ability to observe becomes trained to distinguish among the different types of leaves.

As the children explore on their own, either in books or in the nomenclature material, they are going to discover other characteristics. But those, we don't need to present to them – let them discover them on their own.

After the children have been presented another characteristic, you can take up another activity – simple classification.

October 18, 2004

Simple Classification

Material

The picture cards with no names for the characteristic we are working with – the unnamed picture cards. The tickets that have the names of the characteristics written on them. Various leaves in the environment.



Prologue

The plants the children can classify come from the leaves they have collected or from the plants in the classroom or from other plants in the school.

Get your group of experienced children who have been exploring

Begin

Parallel and Reticulate

Here I have two tickets one says parallel and one says reticulate.

(hold up card)

Where do you think this one goes – into the ? Yes, parallel.

How about this one? Yes, under reticulate.

Now we'll classify these leaves. Here's a leaf. How will you classify this leaf.

(hand out a leaf to each child and let them attempt to classify each one. Lay the tickets on the floor and have them lay the plants in a column under the ticket.)

Epilogue

This is critical because 1) it helps to hone the power of observation, 2) it allows them to use the nomenclature, the vocabulary, 3) it allows them to analyze based on certain physical characteristics.

Because they are doing this activity in a group, there is discussion, debate, defending one's assertion, articulation of a point of view and ultimately a choice is made – how to classify. All of the aspects we have numbered above are the ones that help to develop the human potential.

This exercise is not truly so the children can distinguish between parallel and reticulate. There is a greater purpose at work. So this is another seemingly insignificant activity that is actually huge in its aim.

If a child makes a mistake in the classification, do not correct them. They will correct each other. The right answer is not the important thing here but rather the process.

You can do simple classification with all of the characteristics that you get in biology – not just these two. Make the tickets for this.

This is something that can be done often. They may also want to do a chart or some other activity related. Keep it going; they can go out once a week.

October 18, 2004

Varieties of Leaves Depending on Other Functions

Protection From the Elements

Prologue

This is another direction you can go. Collect a group of children, perhaps a different group.

Begin

Apart from making food, a few leaves have taken up the job of protecting the flower buds. And these particular leaves are called bracts.

They look like they're petals but they aren't; they are leaves.

Epilogue

Dogwood is an example of bracts. Bracts are modified leaves that protect the flower bud.

Protection From Being Eaten

Begin

Some plants have leaves that protect them from animals that would feed on them and they do this in all different ways. Some leaves have stinging cells and they will sting you – like stinging nettle.

Some plants have thorns on their leaves. Holly is an example. Some plants are just bad-tasting. Rhubarb is an example and in fact, the leaf is poisonous.

Protection From Water Loss

Begin

Animals are not the only danger to plants. Sometimes climates with very little rainfall have plants that have adapted their leaves to conserve water.

So these leaves defend against water loss.

Look at the leaf on this plant. It doesn't even look like a leaf (hold up a cactus).

The spines are modified leaves. (hold up example)

This plant holds on to its water so its leaves are juicy and succulent. It doesn't give off its water easily. (hold up example)

This leaf has hairs on it. And when the leaf gives off its water, the hairs hold on to the water for a little bit. (hold up example)

In areas of the world where it is very dry and very cold, plants have prevented water loss by reducing their leaves to the size of needles. In this way, they won't give off too much water. (hold up example)

Epilogue

Hairy leaves is an African Violet. Needles can be any pine or fir.

This is another direction the children can go. We hope they will discover new activities from these we've presented.

Follow-up

Explore the varieties presented in the classroom either outside, in the classroom or in books. Take trips to botanical gardens to see those characteristics not found in the immediate environment. They can press the leaves that they collect, so they can make leaf collections.

They can make artwork with leaves they've collected. Hand-drawing is also

an excellent activity to develop their powers of observations. Working with veins, they can do rubbings.

If they've found interesting leaves that have margins, they can do splatter-painting. (Get a screen, put leaf down on paper, get a toothbrush and spew paint on top of it all and then remove the leaf. The margin of the leaf remains on the paper) Finally, they can use the nomenclature material.

October 19, 2004

The Roots

The Function of the Roots

Material

Rectangular box, drainage material (shards, charcoal, rocks), soil, fast-growing seeds, toothpick

Begin

Put drainage in container, cover with soil. $\frac{1}{4}$ of the way from one end of the container, plant a row of fast-growing seeds. Mark this row with a toothpick.

Water your seeds right on top of the row. Mark the row. Once seeds start to sprout, start to water further and further away from the row of seeds you marked – toward the longer end of the container. Each time you water, mark where you placed your water with the toothpick. There has never been any water put into the short end of the container – this is important. Results are seen days later. Reveal what has happened in the midst of the story.

When you inspect the roots, the root will have grown toward the water. In the quarter pan where there was no water, there were no roots. Water the plant once per day – making sure they water in the correct position.

October 19, 2005

Roots will Grow Around Obstacles

Material

chart E

Setup

Set up the experiment as in the previous experiment – but place an obstacle (rock or large shard) right in the center of the longer part of the pan.

Prologue

Gather you children and tell a story. If the children have had The Leaf, you can say:

Begin

Where does the water come from that the leaf needs to make food? It is absorbed by the roots.

So one could say that the plants eat with their heads and drink with their feet.

If you haven't given the leaf and are beginning with the roots ... say:

How do the plants drink? Well, they drink with their feet. It's the roots that suck up the water from the ground. The water is taken up by the roots and it is sent to the rest of the plant.

Because the roots have the task of finding the water, they have a special sensitivity for it. Wherever there is water in the ground, the roots will seek it out.

Here's our experiment. Water has been placed here. And no water was placed back here.
(point to location with water and without water)

(pull out seedlings and notice the direction in which the roots have grown.)

We can see where there was water, the roots grew toward the water. But we find no roots back here.

Display chart E



It's like you see here. The roots are growing towards the water.

Because **roots take the water from the ground, they rid the ground of its water.** In areas where there are no trees or plants, and it is very rainy, the land becomes swampy – because there are no plants to suck the water out of the ground.

Epilogue

There is a connection between the plant, ground – between biology and geology. This is a cosmic connection.

October 19, 2004

The Parts of the Root

Prologue

It is highly likely you'd continue from the previous presentation right into this one.

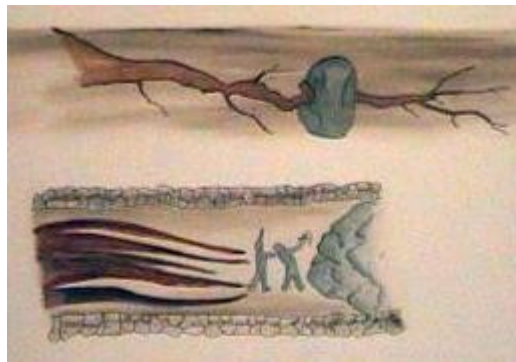
Material

chart F, magnifying glass, corn plant in a clear jar.

Begin

The part of the root that guides the root to the water – searching and feeling its way is called the tip – the **root tip**.

(show chart F)



And as that root tip moves along searching for water, if it meets an obstacle, it will grow right around the obstacle.

However, if that obstacle is too big, the root tip secretes a little bit of acid to break down that obstacle.

So, in a sense, it chips away at the obstacle.

If you cut the tip of the root off, it will lose the ability to find water. So, you could say that the root tip is like the brain of the root.

Because the root tip is so important to the plant, it is protected by a root cap.

(use magnifying glass with a plant in a transparent jar).

You can see a root cap right here, at the very end of the root. The actual absorption of the water is done by the **root hairs** and they are found just above the root tip.

Epilogue

That's the end of the story that goes with the Parts of the Root. It is very easy to see all the parts of the root on a corn seed. All you need for this is a hand magnifying lens and this demonstration.

October 19, 2004

Collaboration between Roots and Leaves

Materials

chart G, H, I

Prologue

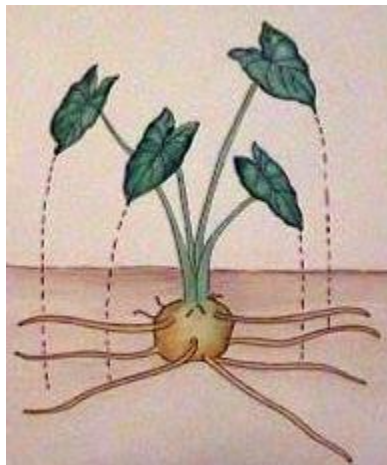
Gather up children who've had The Main Function, certainly they must have had The Root and probably The Leaf.

Beginning

You know, it seems like the leaves and the roots are working together. The leaves somehow direct the water right down to the root tips and the root hairs.

So the roots and leaves work together so the plant can get the water it needs.

(show chart G)



Look here, the leaves seem to be directing the water right down to where the roots are.

(point to chart)

You know plants. They're clever. They never solve their problems in the same way. Look at this collaboration.

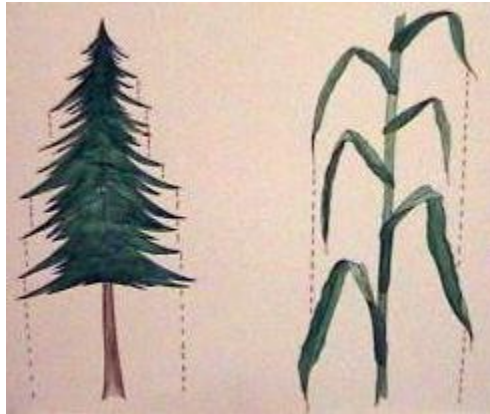
(point to chart H)



This plant seems to point downward from leaf to leaf to leaf to leaf – down to the root hairs.

This one directs its water down the center part of the plant. So its root hairs must be here (point below the plant on the right facing the chart)

(hold up I)



Here's another way. Here's the water – being directed straight downward – leaves down to roots, so that the water is directed right down here to the root hairs below.

And the plants with big heavy branches – usually there is no water directed down here beneath the largest branch at the bottom, so the root hairs are directed right down here (point below the branches to the outer perimeter of the fir tree)

Epilogue

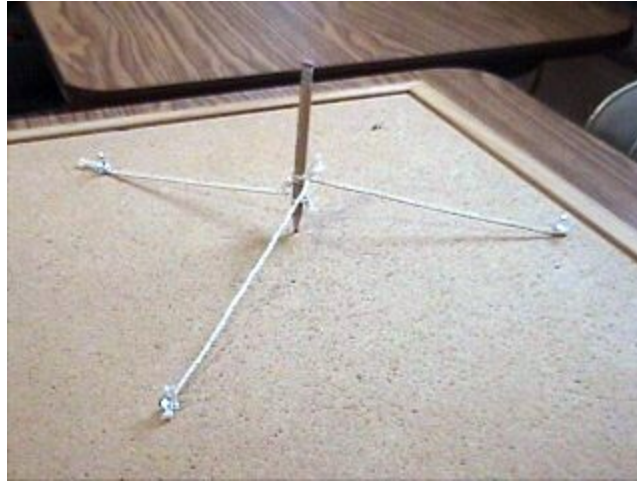
It doesn't matter which is presented first. Chart G makes sense to show first since the arrows show how the leaves are directed to the roots.

October 19, 2004

Roots Hold the Plants to the Earth

Material

Pencil, a cork board, 3 strings, 3 push-pins, chart J



Prologue

At the same time you tell the story, do the demo. Make sure to keep the strings low on the pencil because it looks more like roots which are low on the plant.

Begin

We know that the roots have the job of finding water. And as they search it out, they create quite a network under the ground. As they search this way and that way and this way and that way, they get a very large network.

And this network pulls the plant to the ground. (show chart J) just like you see here.



Look, they're just holding it to the ground. So in a wind storm, the plant won't fall over.

(do demo)

Watch – I'm going to let go. (let pencil drop unsupported)

(attach string to lower part of the pencil and fasten the string onto the cork-board in a fashion after roots)

Now look. It can stand by itself. (pencil stands connected to 3 strings). This is how the root helps the plant stand up.

Epilogue

You can also at the same time take a tiny plant and attempt to pull it up; you can feel how the plant is holding itself in the ground. If you cannot go outside and do this, you can have some little plants growing.

October 19, 2004

Roots Help Prevent Erosion

Material

chart K

Prologue

At the same time as you present, you can pull a little plant up and look at all the soil a tiny little plant will hold onto as you pull it up.

Begin

When the roots are seeking out their water, they create a network under the ground. This network of roots helps to hold the soil in its place too.

In this way, the roots prevent the soil from being washed away by wind or by rain.
(present chart K)



You can see that the roots are holding on to their soil. (point to the roots)

So roots help prevent erosion. (pull up root of small plant)

Look at the tiny little plant – and its holding on top so much soil

Epilogue

Impression comes from the tiny plant holding onto so much soil comparatively.

October 19, 2004

Two Main Types of Roots - Tap and Fasciculate

Setup and Material

Two tall jars to allow roots to grow downward. You need to have blotting paper (can be bought at a stationary store – don't use construction paper. Cut the blotting paper to fit the inside of your jar. A Darker color is recommended because roots are easier to see.

Put the blotting paper on the inside of the jar and fill the center part of the jar with sand. Use sand versus soil because of its consistency and it is cleaner.

Soak the seeds overnight. The kinds needed are corn seeds and bean seeds. Soaking gives them a head start in the absorption of the water.

Once you have soaked them overnight, place the seeds in between the blotting paper and the glass jar side – all around the top part of the jar (around the perimeter toward the top of the jar). It's good to have several seeds because some will mold, some will not germinate – all you need is one.

Once seeds are placed in between blotting paper and glass, water the sand. You can watch the blotting paper take up the water. So moderate how much water you put in. If you put too much water in, your seeds will mold. The last thing to do is to get a piece of black construction paper cut to the size of your jar and attach it to the jar with a rubber band – covering the sides.

Set the jar in a warm place – about 1-2 weeks. Remember that all of the beans go in one container and all of the corn in the other jar. Bottom of jars are transparent – one sees the sand there.

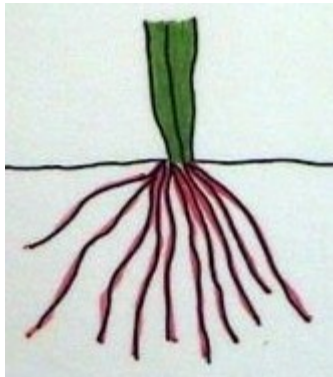
Prologue

Once you give the two types of roots, its nice if you have a garden so the children can go out and explore for these two roots themselves.

Begin

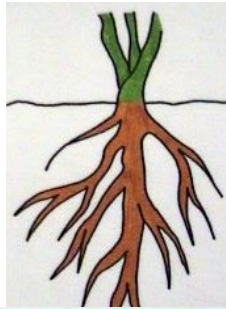
There are two different kinds of roots. This kind of root (point to transparent jar) makes a plant that comes out in a bundle.

We call it fasciculate – a **fasciculate root**. The name fasciculate comes from the Latin word fascis – meaning **bundle**. So roots like these are called fasciculate.



Fasciculated roots are roots which are generally fibrous (threadlike) and all alike. They are included among the adventitious.
(Grass)
(Latin: fiber - thread
fascicle - a bunch, cluster)

This root is different. It just grows straight down. It is called a **tap root**.



A **tap root** is one which
descends deeply into the ground
and generally produces branches.
(Oak, Wallflower, Bean)
(English: tap - to pierce)

Epilogue

The corn is the fasciculate and the bean is the tap.

October 19, 2004

Other Sensitivities of the Root

Roots Dislike Light

Setup

Wire basket lined with sphagnum moss which can usually be bought in garden shops matted together. Into the moss, put a little soil and then put fast-growing seeds all around the surface. Water it and set it in a container that corresponds in size to the basket to keep it dark.

Prologue

If you are excited about roots, the children will also be excited. Make it story-telling!

Begin

Roots are attracted to water. And water is usually found in the ground where it is dark. Where it is light, there is no water to be found. So, in fact, roots have an intense dislike of light.

Look here. (hold up hanging-plant) See these little roots here? I'm going to put them into the light. Watch to see what they do. (hang hanging-plant in the light without the bottom cover)

Epilogue

This is another experiment that happens quickly. The roots turn around and begin to grow back up into the soil.

Best time to tell the story is when the roots are 1/4th to 1/2 inch below the wire basket. The reason is because the longer roots are fighting gravity to go back up and it takes longer – instead they shrivel up and die – not what we want to see.

October 19, 2004

Roots Grow Toward the Ground

Setup and Material

When you set this demo up, make sure to place the beans in all different directions – some sideways, upside down, right side up a variety of position.s

Prologue

As the children do their own explorations, they will make some discoveries. Some of the discoveries will be scientific terms that go with some of your demonstrations. The scientific name for roots growing toward the ground is called **geotropism** – from Greek meaning earth and tropos – meaning **turn**. They go downward regardless of where the water is.

Hydrotropism – Greek for water (Hydro), roots turn toward the water. Also, plants have phototropism – Greek for light (photo).

All these terms may arise related or unrelated to the roots.

Tropism does not mean turning to necessarily – it may turn away from as well and be called phototropic. In this sense it can be said roots are phototropic since they grow away from light.

Begin

Just as roots grow towards the water that is found in the ground, it seems as if they are given the command to grow downward. Right from the beginning, roots will always grow straight downward.

The only time that command is broken is when the water is not found straight below the plant. Otherwise, they only grow downward.

Look. The roots are always growing downward. It doesn't matter which way we place the seed – they always go downward.

Question

Where does the bean plant come from and how are the roots shown?

October 19, 2004

Varieties of Roots

Ariel Roots

Material

If you live in a temperate zone, you may have to look in a book for examples. Real specimens are preferred if at all possible. See underlined.

Begin

In some parts of the earth, where there are tropical forests with plenty of water in the air, some roots have given up the habit of taking water out of the ground.

Under the ground, there is a great competition for the water. So, these clever plants have roots that will take the water out of the air.

(point to example in the book)

These roots are ariel roots and take water out of the air.

Epilogue

Examples are philodendrons, bromeliads and orchids.

October 19, 2004

Root Climbers

Material

See underlined.

Begin

Some roots do other things besides collecting water. Some help the plant to grow up to the sun to get the light it needs. In order to help them attach themselves to other things they have developed suckers.

With their suckers, they can climb up other plants, buildings, even on windows. These are called root climbers (show root)

Here is a root climber. You can see its suckers

Epilogue

Examples are poison ivy, poison oak, ivy and trumpet vines.

October 19, 2004

Pillar Roots

Begin

In some plants when the branches are very very heavy, roots will drop down to support those branches. We call those supporting roots – pillar roots.

(show book)

Here is a plant with heavy branches and pillar roots. Its the Banyan tree which grows in the orient.

October 19, 2004

Prop Roots

Material

See underlined.

Begin

Some plants have roots that are developed from the lower part of the stem – just above the ground.

A row of roots will grow around the stem to help prop the plant up. These roots are called prop roots. In some swampy areas, the prop roots help prop the plant out of the water.

Epilogue

Examples are corn and mangroves, bald cypress.

October 19, 2004

Swollen Roots

Begin

Some of the roots have taken up the task of **storing food for the plants**. These are called swollen roots. There are different kinds of swollen roots.

Here is a swollen root (hold up a carrot) It has a special name; it is called a **conical root**.

Here's another swollen root (hold up an onion) It is called a **napiform root**.

Also, there is a **fusiform**. (hold one up) This is a white radish.

Then we have a **tuberous root** (hold up sweet potato or yam).

These are some of the kinds of swollen roots.

Epilogue

Example of napiform is red radishes and turnips.

Note the three names (napiform, conical and fusiform) are taken from the shape.

Follow up

Have the children explore nature – grocery stores, farmer's markets, outdoors. Sometimes have them investigate neighbors gardens. There is the nomenclature material.

October 19, 2004

Stems

Stem's Main Function

Setup and Material

Fresh piece of celery (old and wilted won't work) with its leaves still on the top. A cutting board and a knife, a glass of colored water (do not make the colored water green). Cut off the tip of the celery (thick bottom tip), place it in the water, set it in the sun in a glass of colored water, watch and wait. It takes maybe one day for the color of the water to tinge the leaves. Choose a celery stalk that has stems growing upward.

Begin

Since the leaves need the sun so that they can make food, and since there is such a great competition on the ground for all plants, the stem has the task of **taking the leaves up to the sun** so they can get light.

(for only those children who have had the root ...)

Just as the root has the command to grow downward, the stem has the command to grow upward. (refer to smaller stems on the celery stalk)

Look how these stems are all growing up!

But the stem has another job as well. It also has the task of **carrying the water that has been collected by the roots up to the leaves** – so the leaves can make the food for the whole plant.

(demo – see above)

We can see how the stem carries water up to the leaves (colored water).

October 19, 2004

Two Main Kinds of Stems

Begin

There are two main kinds of stems. One stem looks like this (hold up plant) It's green and soft and succulent. We call these **herbaceous stems**. Herbaceous comes from the Latin word herba – which means **grass**.

The other kind of stem is called a woody stem. (hold up plant). This is a woody stem (like a fir tree). Plants with woody stems can grow very tall and live for many years. Plants with herbaceous stems tend to be shorter and they have shorter lives.

Epilogue

This is another thing to explore for the children.

October 19, 2004

Parts of a Woody Stem

Material

See underlined.

Begin

We know that the leaves need water to make their food. And we know that the stem carries the water upward to the leaves, that has been collected by the roots.

This water is carried through tubes in the stem. and to protect those vital tubes from either the weather or from animals that would like to eat it, some plants have developed a protective layer around the stem. That layer is called **bark**.

Here's some bark (hold up)

As the stem grows, new tubes are added and the stem thickens. In some plants, these tubes are arranged in **rings**. You can see (hold up small stump with rings) the rings on this piece of wood.

Every year, the plant adds another ring to its stem. And that is how you can tell the age of the tree. **As the tree grows all of the tubes in the center harden and they die.** All of the new tubes are on the outermost part of the stem.

This **hard center part becomes the wood** and that wood is strong but it is also elastic. So, in a wind storm a tree will sway but it will not fall down.

Epilogue

If possible send them out to find trees that have been cut down. You can also look in books to see rings in the stems and the kinds of bark in nature. Keep interest alive by providing them opportunities to explore.

October 19, 2004

Other Parts of a Woody Stem

Begin

You know these vital tubes that are on the inside of the bark, are of two kinds. One kind will send the water up to the leaves. The other kind carries the food downward to the rest of the plant.

Tubes that carry the food are found just inside of the bark. They have the name **Phloem**. Phloem comes from Greek word phloos – meaning **bark**.

The other kind are called the **Xylem**. The xylem are found on the innermost part, closer to the wood. The word xylem comes from the Greek word xulon – meaning **wood**. The **xylem are the tubes that carry the water**.

The xylem is protected by its innermost position because if an animal eats the stem, it will eat the phloem first. But because the xylem is in tact, it can carry the water up so more food can be made. But if the xylem is eaten, the plant can't make anymore food.

In between the xylem and the phloem, there is the **cambium**. The word cambium comes from the Latin word cambiare – meaning **exchange**. The tissue of the cambium can change into either wood or bark.

So these three things, the phloem, the xylem, and the cambium make up the living part of the woody stem.

October 19, 2004

How Water is Moved Through the Stem

Material

chart L

Begin

We know that the stem has the task of carrying the water up to the leaves, so that the leaves can make their food. We also know that the water is collected down in the ground by the roots.

So, the stem sometimes has to carry that water a long long way up. Think about the tallest tree you know – the Redwood tree. Sometimes a redwood tree can be 300 feet tall. And the stem has got to carry the water all that distance to the tip-top leaf.

How does it do that?

The cells of the root are made so that once they collect water the water can't go back. So, those cells collect water and collect more water – and they begin to swell up. Eventually, they can't take any more water – so they push that water to the cell above them.

And that cell takes on water and takes on water, until it can't take on more water. And it pushes its water to the cell above it. And it goes on and on like this.

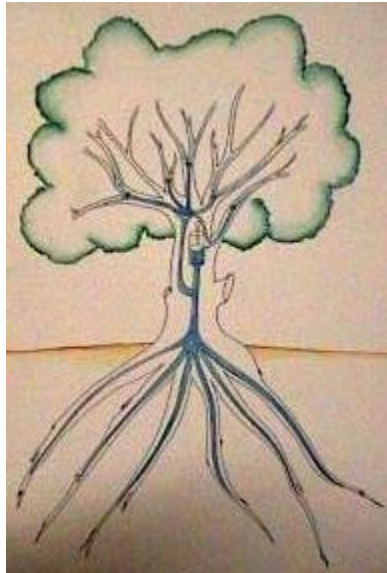
That is called **root pressure**.

But root pressure is not enough to carry the water all the way up. Remember how we saw that leaves gave off their water? When they give off their water, they get thirsty and they want more water.

And so the leaves say, “hey I'm thirsty!” they take the water from the cell below them. Then that cell gets thirsty and it takes water from the cell below it. And that cell gets thirsty and it takes water from the cell below it.

So **through those two things – root pressure and evaporation – the stem is able to carry the water up to the very top leaves.**

(present chart L)



It's like you see here. It's like the stem is a pump, pumping the water out of the ground and all the way up the stem to the leaves. (display chart to side)

Epilogue

Dramatize the story: raise the hands to lift the water and so on.

October 19, 2004

Varieties of Stems

Underground Stems

Material

See underlined.

Prologue

When you cut the bulb, show the main stem in the center so you can find the fleshy leaves.

Begin

Sometimes the animals do get through the stems and they eat the soft juicy parts. Or sometimes, the weather is so bad that it makes life for the stems miserable.

So, some stems have decided to make their lives underground. But we have different kinds of underground stems.

Stems are very clever. This underground stem (hold onion sphere only) has taken up the job of storing food for the plant. And it stores (cut onion with a knife on a cutting board) food in something called fleshy leaves. (misnomer)

We call this a **bulb**.

Other bulbs include hyacinth, narcissus and tulips.

There is another underground stem that also stores food. But the leaves are not fleshy, they are dry. We call this a **corm**.

Sometimes underground stems grow horizontally just under the ground. These are called **rhizomes**. Here is a rhizome (hold up ginger or iris).

And then here is another underground stem that stores food. It is called a tuber (hold up a

potato).

Epilogue

Examples of corms are garlic, gladiolus and crocus. Examples of rhizomes are ginger, iris and begonia. Examples of tubers is a white potato, anemone and ranunculus.

Erect Stems

Look at how straight and tall this stem is. We call it an erect stem. It grows above the ground and grows very erect.

Lazy Stems

There is another stem that does not grow erect above the ground. It is called a procumbent. It grows along the earth.

Strawberries have procumbent stems.

October 19, 2004

Climbing Stems

Material

chart N, M. See underlined.

Begin

Sometimes there are stems that depend on other things. They are also very lazy, so lazy that they climb up on other plants. We call these **climbing stems**.

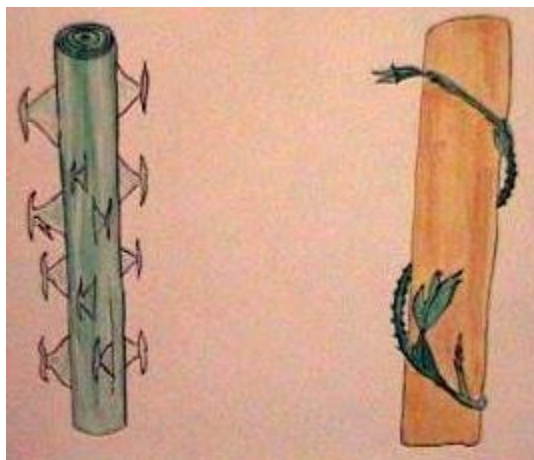
But there are different kinds of climbing stems.

Here's a climbing stem. (present chart M)

It is twining around and growing up. We call it a **twining stem**.

Examples of twining stems: honey-suckle and hops.

Another kind of climbing stem uses hooks. They are called **hook-climbers**. They hook onto something and they climb up. Here (present chart N) we see a hook climber.



Another kind of climbing stem produces a tendril that reaches out and pulls itself up. Here, we see a tendril reaching out.



Examples of tendrils are grapes. Examples of hook climbers are roses and black-berries.

Follow-up

Exploration of the environment beyond the experiences in the classroom. They can examine cross section of trees, study different kinds of woods, identify different kinds of barks. They can ponder why the rings have gaps between them sometimes while very close together at other times. (Lots of rain causes big gaps, else rings are close)

They can study knots of trees. They can dissect onions. Study Sapwood, They can look for all kinds of climbing stems

October 19, 2004

Flowers

Flower's Main Function

Material

It's nice to have a bunch of flowers nearby as you tell it.

Begin

When the flowers are out, the plant is actually calling out. The call is silent but it is very loud. It is a voice of scent and color. And only the insects can hear that voice and understand its meaning. It's saying, "come and get it!"

The plants are displaying their wares. Some are yellow, some are purple, some are orange. There are all kinds of flowers.

And the scents are spread around and it reaches those for whom it is intended. Well, being called – the insect arrives. He looks around and he finds a bit of sweet juice.

That is the nectar that is hidden at the base of each petal of the flower. In the course of his looking, the **insect gets dusted with pollen.** And the plant thinks to herself, "You have come to get my sweet juice but you also get something even better – you get my powder. What a treat!"

The insect is very busy and he goes off in search of more juice. And he will visit another flower. And **if the pistil is ripe on the new flower, it will catch a hold of the bits of pollen** on the back of the insect.

And those insects – **they just go from flower to flower depositing pollen** as they go.

Epilogue

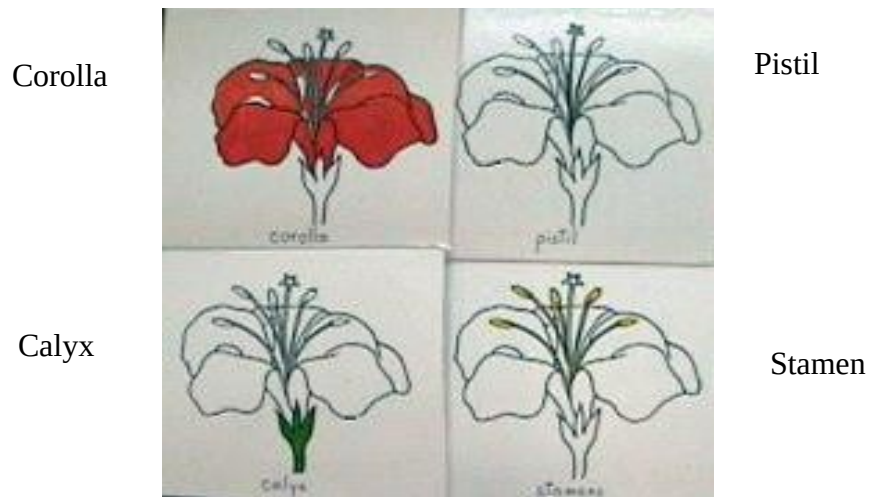
This is a short story for a main function so go directly into the parts of the flower.

October 19, 2004

Parts of the Flower

Material

A flower with 4 parts – a complete flower. Use an Azalea.



Prologue

Children will likely already have some experience with the flower. When you introduce the flower, make sure you have a specimen that has all its parts. You also want to make sure it is large enough for the children to see the parts.

A nice example is the azalea. It is easily gotten and has nice clear parts.

Begin

The part of the flower that **does the calling out** is called the **corolla**. Corolla comes from the Latin word corolla - meaning **crown**.

This is the corolla (show the top of the flower). It is made up of **petals**.

The part of the flower that **protects the flower bud** is called the **calyx**. Calyx comes from the Latin word calyx and means **cup**.

This is the calyx (show green part supporting the petals)

The calyx is made up of **sepals**.

This part of the flower that **catches the pollen** here is called the **pistil** (extended things coming up from middle of petals). It comes from the Latin word pistillum – meaning **pestle**.

The **stamens** are the part of the flower that make the pollen. The word stamen comes from the Latin word stamen – meaning **thread**. They look like little threads. The word **pollen** comes from the Latin word pollen – meaning **fine flour**.

So those are the **four basic parts of the flower**. **The corolla, the calyx, the pistil and the stamen.**

October 19, 2004

The Parts of the Pistil

Material

a pistil

Begin

Here, I have a pistil. The part of the pistil that **collects the pollen** is called the **stigma**. It is the very tip of the middle protrusion. It is usually sticky so the pollen will stick to it.

The part of the pistil that **holds up the stigma** is called the **style**.

This part of the stigma that **holds the eggs** is called the **ovaries**. The word **ovary** comes from the Greek word Ovum – meaning **egg**. The **egg cells are kept in the ovary**.

October 19, 2004

The Parts of the Stamen

Material

A stamen.

Prologue

If the children are experienced with flowers, you can give parts of the stamen, parts of the pistil and so on. If not, its best to pick this up later on.

Begin

Here I have a stamen (hold it up). The part of the stamen that **makes the pollen** is called the **anther**. The word anther comes from the Greek word anther – meaning flower. It is the top of the stamen.

The part of the stamen that **holds up the anther**, is called the **filament**. (this is more or less a stem for the stamen).

Epilogue

Once you have introduced the children to the parts of the flower, send them out to explore. They will find all kinds of interesting things about flowers and their parts. They will discover some flowers have all the parts and some have parts missing. Some flower parts are joined and others parts are not joined. Some flower parts are found in different places than the same part on different flowers.

They will want to note the diversity of the flowers and their parts. This plays generally into diversity within the cosmos.

October 19, 2004

Varieties of the Parts - Characteristics

Material

See underlined below.

Prologue

The children will have gone off exploring and collecting flowers from gardens (with permission). You will also have prepared your own environment so they make discoveries you want them to see.

You'll introduce them to some of the characteristics of flowers and like the leaves, they will be characteristics that aid in scientific classification.

Polypetalous and Gamopetalous

Begin

Look at these two flowers. They're just a little different.

This flower has all of its petals that are free – they are not joined. You can pull them off one by one. We call this polypetalous.

But look at this one. Its petals are joined by a column. When the petals are joined, the flower is gamopetalous.

Epilogue

Examples that are polypetalous are roses, strawberries, apples and azaleas. Gamopetalous are morning glories and gloxinia.

October 19, 2004

Polysepalous and Gamosepalous

Material

See underlined.

Begin

Look at the calyx and its sepals. These sepals are not joined. They are free – attached but free. These are called polysepalous.

In this flower, the sepals are joined. When the sepals are joined, we call this gamosepalous.

Examples of polysepalous is cherries, roses, apples. Examples of gamosepalous flowers are hollyhock and penstemon.

October 19, 2004

Complete and Incomplete

Material

See underlined below.

Begin

Flowers that have all four of their parts - that have the corolla, the calyx, the stamens and the pistils – are called complete.

This is a complete flower (hold up to view)

A flower that is missing one of its parts – doesn't matter which one – is called incomplete. This is an incomplete flower.

Epilogue

Examples of complete are peas, beans, apples, azaleas and cherries.
Example of incomplete flowers are oak, iris, begonia and tulips.

October 19, 2004

Perfect and Imperfect

Material

See underlined.

Begin

A flower that has both its stamens and pistils is called perfect. This is a perfect flower (hold up). It has its pistil and its stamen.

But here is a flower that is lacking its pistil. (hold it up and point). A flower that lack either a pistil or stamen is called an imperfect flower.

Epilogue

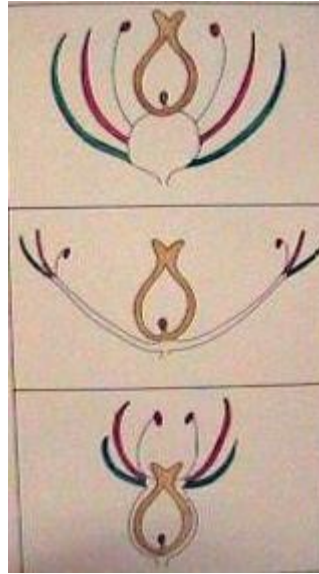
Examples of perfect are lilies, irises, roses and amaryllis. Flowers that are imperfect are begonias, catkins, squashes and pumpkins flowers. Begonias are nice to have in the classroom because on the same plant, you'll have flowers that only have the pistil or only just the stamen.

October 19, 2004

Hypogynous , Perigynous and Epigynous

Material

Chart O



Prologue

The children will be exploring and dissecting flowers. They will cut right into the ovaries and examining where the flower parts originate in relation to the ovaries.

You can show the child a magnolia or a buttercup and say:

Begin

The **flower parts** are **below** the **ovary**. We call this **hypogynous**. The word hypogynous comes from the Greek word gynous meaning **female**. Hypo comes from the Greek word meaning **under**. So the flower parts are under the ovary in hypogynous flowers.

This flower is a little different. It's **floral parts are found around the ovary**. We call this **perigynous**. The word perigynous comes from the Greek word peri meaning **around** and gynous meaning female. Examples are bramble and roses. There are very few perigynous flowers and they are usually in the rose family.

Here we have a flower that has its **flower parts above the ovary**. We call that

epigynous. Epigynous comes from the Greek words gynous, meaning **female**, and epi, meaning above. So the floral parts are above the ovary.

Epilogue

Gynous means female in Greek. Everything is in relation to the female part – the ovary. These characteristics are determined by the rest of the floral parts in relation to the ovary. Hypo in Greek means below – like a hypodermic needle. Hypogynous means below the female part. Peri in Greek means around, so perigynous means 'around the female part.' Epi in Greek means above. So epigynous means above the female part. Examples of epigynous are apples, dandelions, sunflowers and most composite flowers.

October 19, 2004

Specialization of Flowers to Ensure Pollination

Introductory Activities

Collect the carcass of a bee. On the carcass of this animal, point out its hairy back and on its hind limbs – its pollen sacs. The sacs are where the pollen is placed.

If you can find a butterfly carcass, you can roll out its proboscis (tongue). With a straight pin, you can uncurl it and see how long the tongue is in relation to the body.

On a warm day, sitting outside by many flowers, you can see butterflies perch and see the tongue go down into the flowers.

It's nice to have hummingbird feeders outside the classroom. Hummingbird feeders have red juice you put in that attracts hummingbird.

October 19, 2004

Flowers Which Invite All to Come

Material

See underlined.

Begin

There are some flowers which are not very picky. They don't care who eats them. In fact, they sometimes have an arrow pointing the direction to the restaurant. It says, "Go this way. Follow this path!"

This on a flower is called an indicator.

Epilogue

Examples with indicators are pansies, tiger lilies and nasturtium.

October 19, 2004

Flowers That Invite Specific Friends

Material

Picture or specimens of Nasturtium, robber wasp, lark spur, columbine and lupins.

Begin

Sometimes there are flowers that are choosy. They only want special guests.

Here is a flower that only wants butterflies to visit. And the butterfly has to have a long tongue to get the nectar. The nectar is kept way deep in this spur.

Then there is the robber wasp. He gets his name because he robs the flowers. He is not happy with some flowers that only want special guests. So he just chews through the petal to get that precious nectar. He is the **robber wasp**.

There is another kind of flower that knows about the antics of the robber wasp. It has built a protection in its back door to protect its nectar. This is called the lark spur.

Here is a flower that only wants flying insects to visit it. It doesn't want the ant to come in. In fact, to prevent the ant from coming in, it has built up a barricade of fringe so that the ant has trouble getting through it to the nectar. This is the nasturtium – with a petal that has been adapted with a barricade of fringe. Why does he not like the ant? Because he is bald – no hair to carry the pollen!

Then there are some flowers that are so accommodating that they make landing pads for the insect to land on. Once the insect lands on that petal – or landing pad, BOOM, the stamen comes down and strikes or slaps it on the back.

Epilogue

Examples for butterflies only are columbine. Lupins are example that slap the insect.

October 19, 2004

Who Does the Advertisement and How

Material

See underlined.

Begin

Normally it is the flower that calls out to the insect. Sometimes it is not the flower that calls out – it is the leaf. Dogwood and poinsettias are an example of this.

Sometimes its not just one flower but a whole bouquet of flowers. They join together. Look at this bouquet of flowers – they have all joined together. Examples of these are sunflowers, any composite flower and Queen Ann's Lace.

Then there is the flowers that call out at night using scent. Flowers that use their scent and call out at night are usually white. They don't have to have bright colors because white is easier to see at night. Flowers like these are gardenias and honeysuckle.

However sometimes the scent is not that pleasant. Sometimes it smells like rotting meat. Insects still like to visit it – especially flies and wasps.

Epilogue

Examples are skunk-cabbage, trillium, bugbane and stapelia (has biggest corolla of all flowers).

October 19, 2004

Other Ways for Pollination to Occur

Material

See underlined.

Begin

Sometimes its not the insects that spread the pollen around. Sometimes it's **birds**.

Usually the birds like flowers that are large with tubular shapes. Hibiscus is an example.

Then there is the wind. Many plants like to use the **wind** to spread their pollen. It is simply blown from flower to flower. The flowers on these kind of plants are usually small and inconspicuous because they don't have to attract the insets. They usually **come out early in the year**, before the leaves come out on the trees – so the leaves won't become obstacles to the pollen.

The plant is clever in another way. It **makes a profusion of pollen** to ensure that some pollen will reach its destination. Hayfever is caused by the wind carrying the pollen from flower to flower.

Epilogue

There are a lot of stories for ensuring pollination. Don't give them all at once. Tell one or two and hopefully they will become interested and look up other ways that plants have for ensuring pollination. There is an orchid, for example, where the flower parts look like the female version of the bee in order to attract.

Follow-up

There is a lot of drawings by hand or painting that can be taken up with flowers. They can visit botanical gardens, grow their own flowers, go out into the forest and look for little flowers they might not otherwise have thought would be there.

October 19, 2004

The Fruit

Fruit's Main Function

Material

Plate of fruit. See underlined.

Begin

Once **pollen travels to the stigma, which is now ripe**, the pollen is **sent down a tube through the style into the ovary**. A cell from the pollen then travels down that tube to meet the egg in the ovary.

The cell goes down the tube and meets the egg. Then the ovary begins to swell and swell and swell, until it becomes a fruit.

In the ovary, the tiny egg, develops into the seeds of the fruit. And that swollen ovary helps to protect those precious seeds.

The **petals and the stamens** and the other parts of the flower have done their job. And so they just **wither and dry up** and drop off.

(pick up apple)

Here on this fruit, you can see the dried stamens (point to the stem-like woody narrow growth that enters at the top of the apple, growing into the center of the apple)

On this fruit you can see the dried calyx (point to the woody narrow growth that enters at the top of the pepper, growing into the center of the green pepper).

Look here. On this fruit, you can see the stigma and the style (small hair-like thing on the end of a pea pod).

Here we can see the seeds (cut open green pepper). They were once seeds inside the ovary.

Because the fruit is so good to eat, some plants have devised creative ways to protect themselves.

This fruit grows underground (show a peanut)

This fruit has a very prickly outer covering (chestnut).

This fruit (coconut) makes a hard covering so the animals can't get to the seeds easily.

The **fruit is what protects the seed.**

Epilogue

Cut open a green pepper to see dried up seeds.

The **pollen after attaching to the stigma creates two little cells. One cell creates the tube and another travels down the tube.**

October 19, 2004

Two Main Kinds of Fruits

Material

See underlined.

Begin

Here I have two fruits (apple and peanut). This fruit which is very juicy and succulent is called a **succulent fruit**.

This fruit is not succulent. In fact, it **has a very dry covering**. We call this a **dry fruit**.

Epilogue

Examples of dry are hazelnuts, acorn, peanuts, any grain such as rice, grass, oats.

October 19, 2004

Parts of a Succulent Fruit

Prologue

Make sure to have a succulent fruit that has all of its parts. Cucumber and a seed-containing plum (not seedless) are good examples.

Begin

Here I have a succulent fruit. The whole thing has the name **pericarp**. Pericarp comes from two Greek words, peri which means **around** and carp from the Greek work karpos, meaning **fruit**.

See this one? This part is called the **epicarp**. (cut skin off and raise to show) Epicarp comes from two Greek words epi which means **above** and carp which means **fruit**. We also call this the fruit's skin.

This succulent part in the middle is call the **mesocarp**. Meso is a Greek word meaning **middle** and carp, again meaning **fruit**.

This part here (hard part on the inside of a plum) is called the **endocarp**. Endo is a Greek word meaning **within**. Within the endocarp, there is a seed. The **endocarp protects the seed**.

October 19, 2004

Kinds of Succulent Fruits Based on the Parts

Material

See underlined.

Begin

When you have a **succulent fruit that has a soft epicarp and a succulent mesocarp and a hard endocarp – and it surrounds one seed – this kind of fruit is called a drupe**. The word drupe comes from the Latin word **drupa**, meaning **over-ripe wrinkled olive**. Plums are an example of this as well as a cherry.

When you have a fruit that has a **soft epicarp, a succulent mesocarp and it surrounds a whole lot of seeds, you have a berry**. This is a berry (holds up a tomato). Green pepper is another example (hold open a cut green pepper)

Examples of these also include grapefruit, green pepper, tomato, kiwi and orange.

When you have a fruit that has a **soft epicarp and is very closely attached to a soft mesocarp so that it forms a rind, and it surrounds lots of seeds, we call it a pepo**.

Other examples are pepos are zucchini, cantaloupe, pumpkin.

When you have a fruit that has a **soft epicarp and a succulent mesocarp and it has a core that has about 4 or 5 seeds in it, we call that a pome**. Examples are apples and pears.

Epilogue

The receptacle (which holds up the ovaries) on an apple grows and grows. The ovary is the core. **It is the receptacle that we eat**.

Children love to paint fruits, draw them and so on.

October 19, 2004

Kinds of Succulent Fruits Based on the Flower

Begin

When you have a **fruit that comes from a single flower and a single pistil**, you have a **simple fruit**. This is a simple fruit (hold up tomato). It came from a single flower and a single pistil. Most succulent fruits are simple fruits.

When you have a **fruit that comes from a single flower with many pistils, the pistils groups together to form the fruit, you have an aggregate fruit**. This is an aggregate fruit (hold up strawberry). Aggregate comes from Latin word aggragare, which means to **group together**. Raspberries is another example.

Each one of the things we call seeds (little outer things on a strawberry) is a separate fruit and the red part we eat is again the receptacle.

When you have a **fruit that is made out of many flowers, you have what is called a composite fruit**. Here we have a composite fruit (show a pineapple). All the little facets on the pineapple was a flower. Other examples are mulberries, figs and an Osage orange (also called a horse-apple).

October 19, 2004

Kinds of Dry Fruits

Begin

When you have a dry fruit that bursts open and throws out its seeds, it is called a dehiscent fruit. Here I have a **dehiscent** fruit.

A dry fruit that does not burst open and throw out its seed, is called an **indehiscent** fruit.

Epilogue

Examples of dehiscent fruits are poppy, peas, violas, milkweed and dandelions. Examples of indehiscent are maple, elm, magnolia, mustard, radish and iris.

Follow up

Explorations in books, grocery stores, farmer's markets. In the neighborhood, looking at bushes, there are tiny fruits all over. Books are good places for exploration. "From Flower to Fruit", by Ann Ophelia Dowden, is an excellent book to have in the classroom.

October 20, 2004

The Seed

Seed's Main Function

Begin

The **seed is the part of the plant that houses the new plant**. The baby plant lives inside of the seed.

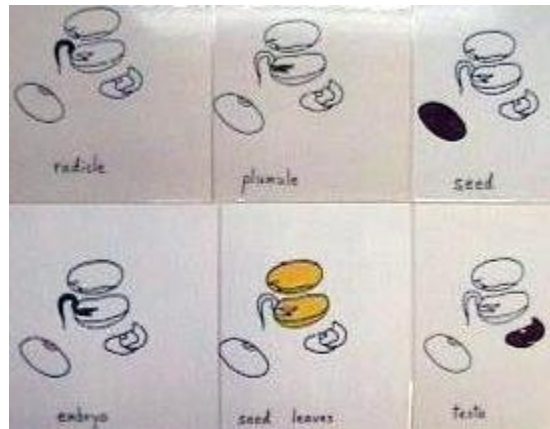
But not only does the seed house the baby plant, it also **provides it with its food**.

October 20, 2004

The Parts of the Seed

Material

See underlined for examples for the classroom.



See below for description of parts

Prologue

Have some specimens of seeds that have been soaked overnight. The best is a lima bean. Take the well-soaked seed and try to separate it from the protruding ridge – opening it very carefully. If you open it very carefully you will see the baby plant.

Begin

Here is the baby plant. We call it the **embryo**.

The embryo is made up of two parts. This part (lower section without hairs) is called the **radical**. Radical comes from the Latin word radice – meaning **root**.

The **radical is the part of the embryo that will become the root**.

The **top part with the hairs, is called the plumule**. Plumule comes from the Latin word pluma – meaning **fine feather**.

These parts of the seed are called (hold up the two outer containers of the lima bean) **cotyledon**. They come from the Greek word **cotyledon** – meaning **cup-shaped cavity**.

The whole **seed is protected** by this part, called the **testa**. The word testa comes from the Latin word testa – meaning **shell**. Like a shell, it is very hard.

Epilogue

You should have for them **soaked overnight all kinds of different seeds** they can explore. As they explore, they try to open them up and will discover that some seeds open up easily and others do not open at all. This is when we introduce the two kinds of seeds.

October 20, 2004

Two Kinds of Seeds

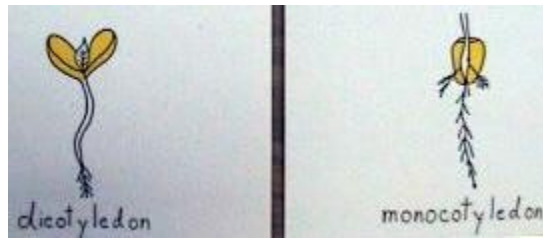
Material

Chart P, two seeds – one monocotyledon and one dicotyledon.

Begin

This seed, which has two cotyledons we call a Dicotyledons. The word **Dicotyledon** comes from two Greek words. Di – meaning **two** and cotyledon – meaning **cup-shaped cavity**.

This seed, which **you can't break into two parts, we call monocotyledon**. Monocotyledon comes from two Greek words, mono- which means **one** and cotyledon – which means **cup-shaped cavity**.



Monocotyledons always have **leaves with parallel veins** and they will always have **fasciculate roots**. They will always have **flower parts in threes or in multiple of threes**

Dicotyledons always have **leaves with reticulate veins and tap roots**. **Flower parts are always in fours and five or in multiples of fours and five**.

They both will always stay true to their tribe. A Lima bean seed will always turn into a Lima bean plant and a corn seed will always produce a corn plant.



It's like you see here (point to chart). The mother plant is sending off her baby plant and he has his one sack of food. And she says to him, "Go my child and remember that you belong to the noble descent of the monocotyledons! Where you go, maintain our tradition."

Epilogue

Sometimes cotyledon are called **seed leaves** or seed coat. Give the children time to explore and experiment. Provide them an opportunity to plant some of their seeds. Allow them to **plant all monocotyledons in one container and all of the dicotelydons in another container**. Make sure they label the gardens. Monocotyledon include lilies, palms, grass. Dicotyledons include strawberries, oaks, maples, apples, peaches and willows. (Most plants are Dicotyledons)

After the gardens sprout, point out, if they don't discover it, the following:

- monocotyledons always have leaves that have parallel veins
- monocotyledon always have roots that are fasciculate
- Dicotyledons always have leaves that have reticulate veins
- Dicotyledons always have tap roots

Allow the garden to continue to grow. If they are nurturing and caring for them, they will get some flowers. When they get flowers, they will notice the following:

- monocotyledons always have flowers where parts are in threes or in multiples of threes
- Dicotyledons always have flower parts in fours and five or multiples of fours and five

Once the children have experienced these facts for themselves, summarize for them with chart P (see last part of dialog)

October 20, 2004

Simple Classification

Material

A number of plants and certain cards from the nomenclature material (the unnamed pictures of the two kinds of seeds, the unnamed pictures of the two kinds of veins systems in the leaves, the unnamed pictures of the two kinds of roots). To serve as heading cards, use the tickets for the two kinds of seeds (Monocotyledonae and Dicotyledonae).

Setup

Layout on the floor the two tickets (monocot and dicot) as heading cards. Place the two pictures of the seeds. Place either the leaves next or the roots nomenclature cards. Plants are then classified under all cards into columns of plants placed on the floor.

Epilogue

The activity here we want is discussion, debate and defending one's assertions. This provides critical thought. Let the children have their own discussions without correcting.

October 20, 2004

Seed Dispersal

Material

Chart P, chart Q. See underlined.

Prologue

Remind the children that the mother plant sends her baby plant off into the world but sends him off with protection and a sack of food. The reason she must send him off is because there is a problem with plants. If the baby were to settle down near the parent plant, it would have a hard time making its food. The parent plant is large and will probably take up most of the sunshine and most of the water. It's not like with animals; they like to settle down with their parents – but animals can move and the plants cannot.

So the plants have devised many clever ways of sending off their babies.



Begin

This plant offers the gift of food to the animal who will carry it away. (hold up strawberry) If the seed is too large for the animal to eat, he will just devour the fruit and deposit the seed elsewhere.

However, if the seeds are small and indigestible, the animal will eat the whole fruit – seeds and all – and then deposit later, somewhere else with its own supply of manure. We call this **Gift of Food** seeds.

Then we have a plant that makes a fruit that will stick to some animal and the animal will carry it off (stick the spurs to yourself) and later scratch it off. Those are called

hitchhikers.

But then, we have some plants that when their fruit is ripe, it just explodes. And it sends its seeds off. Those are **exploding** seeds.

You know sometimes the plants make use of the wind. This fruit (hold up one) has a wing to carry it off away from its parent. We call these **winged** seeds. And this one (drop it) has a parachute – we call it a **parachutist**.

Then there is the plant that makes a **seed that can float**. It will float and float and when it reaches the beach, it will take root. We call these **floaters** seeds.

(show chart Q)



It's like you see here. Here you have the seed that goes in the front door, is taken somewhere else and goes out the back door. Here, you have the seeds that make use of the wind – the parachutists and the ones with wings. And here we have the exploding seeds. Here we have the floating seeds. And this one is the hitchhiker. (point to each one)

Epilogue

Hitchhiker examples are teasels, stick tite, burdock, cocklebur and puncture weed. Exploding seed examples are impatiens, witch-hazel, pansies, iris and monkey's-dinner bell (when it explodes, monkeys hear it and come to feed). Parachutist examples are sycamore, cattails, milkweed, cottonwood, dandelions and tumbleweed. Winged seed examples are maple elm, ash and box elder. Floating seed examples are coconut, American lotus, water lilies.

After the children become aware of seed dispersal, let them find examples in

books and outdoors. One book is “Seeds, Pop-Stick and Glide”, by Patricia Lauber. It has all different ways seeds are dispersed and is a classic.

Follow up

They love to draw seeds. They use seeds to decorate their work. They can make collages – crop art – out of seeds. They enjoy chart Q a lot. They make the chart and paste the seeds onto the chart. They enjoy writing poetry about seeds and seed dispersal. Finally, they can explore books. This should be part of the prepared environment.

February 22, 2005

Lecturer: AMI Elementary Trainer, Phyllis Pottish-Lewis

Introduction to Zoology

In the Casa Dei Bambini

Children are aware of animals before they come to school because there are so many in their environment. What is interesting about animals to the young child is that they move about and they make sounds. They attract our attention. If the children don't have animals in their homes – but they see them outside in nature, in parks and they see them in zoos as well.

When the child moves to the Casa, his experiences with animals should be extended. If the environment is prepared, there will be animals in the primary class. When there are animals in the primary class, the teachers can direct the observations of the children. What the child sees is that animals move, need food and water to live, different animals eat different things, their cages need to be cleaned, different animals have different behaviors – some are shy, some are active – others are not. They learn the parts of the animals bodies – from the card material. And in their investigations about animals, they learn different facts about their lives.

Animals are another means for helping the child develop their human potentials. Due to their interest in animals, vocabulary is developed. In reading and writing about animals, their reading and writing develop. There are many opportunities for self-expression regarding animals. They are developing powers of observation as they watch animals. To a certain extent, they are building responsibility as they care for animals.

Since all activities in the primary class are designed to help develop the human potential, we can understand that we are not teaching zoology as such. Having animals in the classroom has a greater purpose – beyond zoology.

In the primary class, the teacher directs the children towards the needs of the different animals. This approach continues in the elementary class as well. Just as we directed the child's attention toward the needs of the plant, we now direct their attention to needs of animals.

The elementary child no longer has the absorbent mind. It is now a mind that reasons and that mind is aided by the imagination. We use these two particular traits when we approach study of animals. We try to impress on the child the cosmic aspect of all of the life on the planet cooperating / co-existing with animals - and vice versa. The child comes to know that animals have a relation to the earth and its other inhabitants.

Animals have to meet their survival needs and as a result of meeting their needs, they develop a certain behavior. With animals in the classroom, the child can soon learn that the behavior of animals is directly related to satisfying its needs for survival. Another important aspect we want the child to learn is that there is a great variety of features that results in the animal world. Animals are meeting their needs in different ways. We want our children to understand that animals are adapting physically to the different environments in which they live.

The elementary child is interested in the how and why and they apply this to animals as he does with all else.

Things the Teacher Should Know

In order to direct the child's observations, the teacher must know something about animals. He will use the needs of the animals as the structure for presenting zoology.

What do animals eat? How an animal satisfies this needs gives a clue to its physical features and behavior. If we think about features, we can think about the deer, for example, who eats grass and bark. So, the teeth of the deer are designed for gnawing and grazing. The lion is a meat eater. So, its teeth have been designed for tearing and cutting. The two different diets require two different kinds of teeth – resulting in a variety of features for each animal.

The behaviors of the animals differ too. The lion is a predator. He goes on the prowl when he is hungry; he is aggressive and a stalker. The deer is a plant eater and so he meanders from tree to tree. But all the time, he is very aware and watchful for danger. From these two examples, one can see that what an animal eats influences his features and behaviors.

Another aspect to know about is how animals move. Birds that fly have much lighter bones than animals that walk on the earth. Their bones are porous, not dense – so the bird can support its weight when airborne. Birds that do not fly have much heavier and larger bones than those that fly. In fact, the bones of an ostrich are so strong and powerful that one of its kicks can kill a human being.

Birds also have prominent breast bones. Those breast bones allow for strong flight muscles to be attached to the sternum. Animals living on the ground don't require large chest muscles.

Another aspect of interest is where animals live. This influences a variety of features in animals. If we think of apes living in trees, one finds they have longer arms and lighter bones than apes that live on the ground. A mole that lives in a tight burrow underground goes forward and backward in his little burrow; so his hair on his back is designed so it too can change direction when the mole reverses direction. The sloth hangs in a tree for a week at a time upside down and so its hair grows on its body, the ventral side of the body and hangs down. Most animal's hair is the dorsal side of the body,

hanging down. It hangs this way to discharge the rain and anything that falls on it. Unlike every other mammal that has only seven cervical vertebrae, the sloth has nine; this means the sloth has a greater range of motion for its head. Because it is one of the slowest moving animals, they can't run from its predators. So its range of motion is well-needed.

Presenting the Animals

First of all, we excite the child by making them aware of the immense variety of animals. Once interest is started, we leave them free to explore whatever has peaked their interest. Once they begin exploring, they make discoveries and in turn, the child is motivated toward more exploration and discoveries. So, they continue to build a store of information that they can use as they develop their knowledge and intellects.

We said the structure for presenting animals is based on their needs. For primary, we said there should be animals in the classroom. When this is followed, the children learn from first-hand experience with the animals. If we take the need for food, the child can see that animals need to eat though they each eat certain things. Snakes usually eat only live mice – they're not interested in dead, cold mice. Silkworm larvae are only interested in the mulberry leaf – not the maple or rose leaf. The children become aware of how the animals eat. Snakes unhinge their jaws and eat their food whole. Rats nibble at their food – bits at a time. Tarantulas inject their food with a poison and dissolve eat – sucking it up afterward. Snails eat yet another way.

When you have animals in the classroom, the child becomes aware of different body parts. Birds have four limbs but two of them are modified for flight. Spiders have eight legs; they are jointed and this allows them to move quickly when needs be. Frogs have four limbs but their hind limbs are more muscular and powerful – permitting them to jump long distances. Snakes have no limbs, no legs, but still move by slithering.

How do animals breathe? If you have fish and tadpoles in the classroom, the children see they live in water. If doing dissections, the child will see gills these animals breathe with. An axolotl, an amphibian, always has external gills. Birds and mammals breathe using lungs and grasshoppers breathe with their tails; they have little holes in the abdomen of their bodies called spiracles.

Another aspect for presentation is how different animals sense the world. The fish has no ears but it can still hear; it senses vibration through a **lateral line**. It is a horizontal line from its eye backward to its tail. A snail uses an antennae. Snakes smell with their tongues; this is why it is always going in and out. Pit Vipers, such as a rattle snake, have heat sensors in their cheeks.

What kind of homes do animals have? Birds build all kinds of different nests. Wasps have nests also but they are different – though different from those of birds. Snails carry their homes on their backs. Termites live in wood.

How do animals produce their young. If there are animals in the classroom,

children can naturally become aware of this aspect. Frogs, snails and fish lay many jelly-like eggs. Birds and reptiles lay eggs with hard coverings – though there is a difference in kinds of hard coverings. Birds have a hard shell while reptiles, including snakes, have a leather-like shell. Rats and mice in the classroom give birth to live young. Some animals go through a metamorphosis. 'Meta' is body and 'morphosis' means to change. Tadpoles change into frogs and silkworms become moths. There is a difference between a moth's cocoon while a butterfly has a chrysalis.

Teacher's Responsibility

The teacher needs to know about animals in order to direct the child's exploration. There must be a method by which the child can go out and explore.

The responsibility of the child in the classroom, for the animals is also important. The animals need to be cared for well. They must be fed and watered regularly. Their cages need to be cleaned regularly. This is not the responsibility of the teacher – it must fall to the children. However, the teacher must set up a system to prompt the children to care for the animals independently. By caring for the animals, the child develops a sense of responsibility for other living organisms.

Ultimately, the teacher is responsible for each living thing in the environment – and must be ever observant of animals in the classroom. One of the ways a teacher can make sure the children take care of the animals is by **providing a book for each of the animals** in the classroom. At the beginning of the book, needs are listed – what they eat and how often. Watering instructions and cage cleaning is there as well. The children should get into a habit of keeping this log – with date, food given, how much and who gave the animal care. If anything interesting happens with the animal – spider molted, a celibate bird lays an egg – anything event out of ordinary should be noted in the log. This is a way of helping the teacher track what the children have done and to hold them accountable.

When the children care for the animals themselves, they learn what to feed them and how often. Overfeeding fish can kill them – so the child learns important information. Snakes or other reptiles won't eat if they are too cold or are molting. The child learns that birds need to have grit put into their food since birds have no teeth; normally small rocks are included to break up the food. Amphibians have to have their food moist – so their food needs to be misted daily. If there is a Chinchilla, there must be a way to take a dust-bath. They do this to kill parasites with spiracles for breathing.

When there are animals in the classroom, there are many opportunities to teach respect to the children. Along with respect comes consideration – things the child must learn through experience. One of the ways they do that is by cleaning the environment of the animal. They should do this with the least possible disturbance to the animal possible. How to handle animals carefully is also critical. Give a lesson each year on how to handle each animal.

Respect and consideration is also shown in knowing when to approach an

animal and when not to. Mammals caring for their babies do not want to be approached. Snakes that are molting do not want to be held. Make the children aware that animals need to be cared for year-round; so, who cares for the animals during vacation?

Getting Animals into the Classroom

It's possible one is in a school without a budget for animals. So how do we get animals into the classroom? Many children like to bring in their pets for a short time. Or, one can rotate animals from class to class. Sometimes, one can find lending museums which will lend out a rabbit or similar animal for a period of time. Sometimes, we can find some animal in nature; in this case, be diligent to create the proper environment for its survival. At some point, this animal should be returned to exactly the same place from which it was taken.

If you have no experience with animals, this does not mean there should be no animal in the classroom; the teacher can learn as well. Teachers can share knowledge. Parents can help us learn and sometimes the children can help. Begin on a very small scale and use books to increase knowledge. Choose animals that are hearty varieties indigenous to the local environment; frogs, snails or gophers are often good.

If you have experience with animals, take care to not make the classroom into a zoo. Use a guideline having **one of each class of vertebrate: mammal, bird, reptile, amphibian and fish**. Also, have one or two invertebrates if possible. Let go of birds and mammals if scaling down since the child sees birds and mammals regularly. Good birds to have are cockatiels. They love human company – though they are messy so one should keep a towel.

Be conscious and **careful with children with allergies and fear**.

February 22, 2005

Story Material

Purpose

Show that animals have needs and they are met in different ways. Make the children aware of animals outside their immediate daily experience – such as exotic animals. **This material is used for simple classification of animals.** It makes them aware of similarities and differences and provides a foundation of facts.

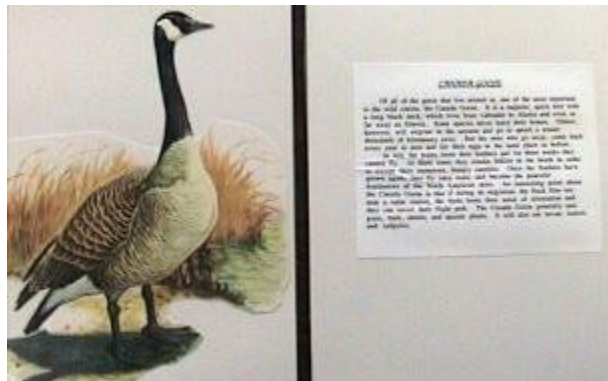
Follow-Up

Explore different animals outside the classroom. Look into books to discover new facts and create reports using them. Go out to explore marshes, zoos, animal reserves, locations of certain activity, such as nesting. Make lists of animals, based on the question and answer cards, sometimes drawing pictures.

Material

Story Folders

Hand-made folders (with content) which are sometimes difficult to find all parts. **Each folder has stories about one particular type of animal, with a variety and sample across the vertebrates and invertebrates.** There should be different folders of contrasting animals – not all mammals for example. Most of the folders should be from animals in the immediate environment. There could be one or two that is exotic. One needs perhaps **four or five stories.**



Large picture and text cards (2) that introduce the animal

Try to have **both vertebrates and invertebrates**. On the outside of the folder, there is only the name of the animal – no features. Transparent folders are nice if you can find them. **Inside the folder one finds:**

- **2 large cards, one has a picture** of the animal in its environment – clearly depicting its features and the **second is general information** in text format about that animal.
- **Smaller picture cards** (8-10inch) that pick up **details** about the animal's life such as reproduction, food gathering, defense and so on. Each card has a corresponding text card.
- **Even smaller picture cards of the food the animal eats**. Each of these cards also has text cards with names of the food. The purpose of these are for children who cannot read well yet – but want to participate in a less challenging manner.



8 x 10 inch picture cards and accompanying text cards



the animal with small picture cards of what the animal eats

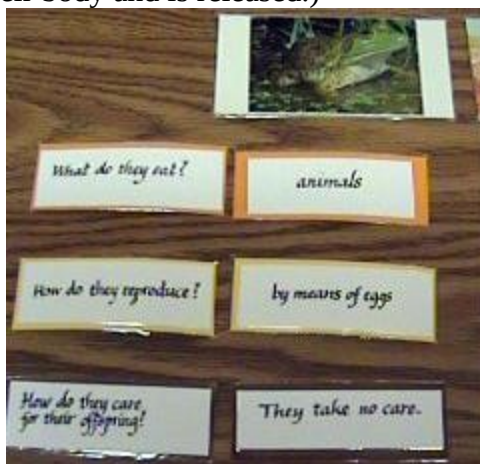
Question and Answer Cards - Classification

Something used with the material but not in the folders are **question and answer cards**. Each set has a series of a question written on its own card and another card with the answer. Such as: **How do they care for offspring?** (suckle, provide food, take no care). The questions and answers are color coded.

Questions (on different colors) are:

1. **By what means do they move?** (creeping, swimming, walking, flying, other means).
2. **What do they eat?** (animals - carnivorous, plants-herbivorous, animals and plants – omnivorous).
3. **In which element do they live?** (in the air – ariel, ground-terrestrial, in water-aquatic).
4. **How do they reproduce?** (born alive-viviparous, by eggs-oviparous, other means-ovoviviparous).
5. **How do they care for their young?**
6. **How are they protected?**

(Note that ovoviviparous animals tend to be amphibians with eggs held in the stomach until the eggs breaks in their body and is released.)



animal pictures continue right in columns, question and answers continue down in rows

These questions and answers form the basis for classification.

There is also a set of pictures the children are already familiar with – typically taken from the primary class and which are known by name.

Prologue

Give this in the early days of lower elementary. It is possible the children have seen all from primary class – except for the question and answer cards. The children work through this material quickly. It is a transition resource because after interest is initiated, they will want to explore animals of choice – not only those in the Story Material. If there are many sets made, rotate them.

For a 9-12 teacher, you probably won't have this material.

Emphasize needs of each particular animal. It is useful in connection with their experiences with animals in the classroom.

Begin sitting together, with an oral introduction with a group of children – each with a folder, including the teacher.

Oral Introduction - Circle Discussion of Animals - 1

Begin

(describe the life of the teacher's animal, modeling the kinds of things we can speak about – food, environment, locomotion, reproduction, defenses and something noteworthy)

(give a child an opportunity to tell a story about his animal – this knowledge comes from his own experience based on the local environment)

(ask questions to stimulate or help each child as needed)

Epilogue

This is a **simple activity that begins discussion about animals in general.**

February 22, 2005

Reading - Working Toward Books Through Needs - 2

Material

Large picture card, large text card, smaller **needs** picture cards with corresponding text cards.

Begin

(use a **picture card as the heading card** under which we lay other corresponding cards)

(have child look at the large picture card and speak of their characteristics and environment)

(invite children who read well to **read the large text card**)

(place large text card next to the large picture at top of workspace)

(use smaller picture cards to lay out in a horizontal line and let the children examine them)

(have child **read a random text card** – placing it under the corresponding picture)

Epilogue

Once a folder is presented, allow the children to work through the others independently. Don't present all folders as above. This is a piece of transition material so this may initiate the child's search for other information in books. They may read more on this or move into other related areas of study.

Note this material and activity is a link between biology and language. This may come up in exams.

For reading practice, food cards can also be used with this presentation along with the prepared labels for the food.

Story notes: born from an egg, instinctively heads for water at birth, nick-named battering ram, quite ferocious, sharp-edged mouth that can bite through bone.

February 22, 2005

Working Toward Books Through Food - 3

Material

Large picture cards, large text cards, smaller **food** picture cards with corresponding text cards.

Prologue

By this time, the children have had experience with the folders and know some facts about these animals.

Begin

(take all large picture cards and mix them up)

(classify food cards according to the food each animal eat)

(food cards appear above the large picture cards)

(use simple tickets for children who don't read well, having the child place the tickets next to the food cards)

Epilogue

They can read this material aloud – though it is a limited challenge. This is quite different than having them read from a literature book. Be sensitive on this.

February 22, 2005

Using Question and Answer Cards - 4

Material

Question and answer cards, box of animal picture cards.

Begin

(lay out question alone as a title)

(underneath, lay out the possible answers underneath that question)

(since they've had experience with these animals, they often know the answers)

Here's a moose. (hold up picture card) What does he eat? (place picture under answer card)

(repeat for other animals)

Epilogue

The **main question is a header. Under are possible answers as other headers in columns. Underneath the answers, lay the picture cards of the animals that apply** to that category.

They will discover that different animals eat other animals, others eat plants and so on. After the analysis and classification is made, give them etymology of words if they appear on the back of the answer cards. "Carnivorous means 'meat' (carni) and vorous means 'to feed upon.' Herbi' means plants and 'Omni' means many in Latin.

After this, **vary the exercise by choosing an animal** and asking what they eat. Then ask how they care for offspring. In other words, use the question and answer cards by animal – as opposed to the lesson which uses the question and answer cards according the category, then assigning the animals.

Parallel Activities: Body Function Awareness

Prologue

These activities draw the children's attention to certain characteristics of animals. Whenever we point out characteristics of an animal in the elementary class, we always want to relate it to the animal's life in the environment.

This means we relate the features of the animal to the function the feature performs. The features we introduce the children to are usually ones that help in scientific classification.

Body Coverings

We want to draw the children's attention to the fact that fish have scales and typically the scales go in one direction and overlap layers. The reason they do this is so they can move through the water swiftly without unnecessary drag. Point out the scales on a real fish – in the aquarium or from one taken from the market.

Let the children know there are different kinds of scales. They may want to look into the different patterns – particularly sharks. **Sharks have an ancient scale** to them. Point out the shape of the fish's body. It's called a **fusiform** shape. The **scales on the fish are moist – making a distinction between scales on a reptile and scales on a fish.**

When speaking of amphibians, point out they have a moist skin and this is why their environment must be misted regularly. **Amphibians actually drink through their skin.**

With birds, we want to draw their attention to the **three kinds of feathers: primary, secondary and down.** Primary feather is used for flight. The secondary feather is used for stability and down is used to keep the animal warm. Then look at feathers in particular. The bird's feathers are designed so they can lock together, so they are smooth and wind-resistant. So, there are locking arrangements called **barbules**. You can see them carefully and it is said this was the inspiration for zippers. They use their beaks to lock up their barbules.

Then there are **mammals. They have fur but they have a different quality to the fur- ranging from very coarse to very fine.** Sometimes lending museums will lend stuffed animals – a mountain lion or other animal normally not accessible becomes accessible.

Respiration

Make the children aware that all animals breathe and that means they take in oxygen in some manner. Animals that live on land take in oxygen by means of lungs. Animals that live in water use gills to breathe. If one does a dissection of a fish, one should examine carefully the gills.

Skeleton

Make the children aware that many animals have a backbone. It provides **support for the muscles and it gives mobility** to the animal. Point out that a backbone is like a cylinder but is broken into segments and these segments allow for movement.

If one has a tube that is not divided and another that is divided, it can be shown what mobility is gained in dividing it into segments (empty paper towel dowel). You can look at the children's backbones as well – done respectfully.

If you want to drum up interest, take an animal from a butcher's shop, boil the animal and soon all of the flesh will drop from the bones – leaving a first hand revelation of a complete skeleton.

When you see a complete skeleton, point out the **most massive part is the skull**. This is because the skull protects the brain and the brain is quite important to the organism. Also point out the **thoracic cavity**. It has **more bones around it to protect the heart and lungs – also quite important** to living creatures.

Circulation

We want the child to become aware that the **heart is a pump** and that it pumps blood all throughout the body. It circulates it throughout the body. As a unit it is called the **circulatory system**. The children should be made aware that the heart starts pumping even before the animal is born and it will pump every minute of that organism's life until its death.

Have a **stethoscope** in the classroom so the children can listen to their own hearts as well as to some of the animals in the class. Listening to a bird can be challenging but others animals are easier.

Look under the microscope at a fish's tail because you can actually see the red blood cells pulsating in the tail. There are **arteries, veins and capillaries**.

Gold fish are hearty so use it – catch it in a net. Wet hands first to avoid damaging its scales. Wrap the fish in a wet paper towel – not the bare hand. Put his tail

under the microscope and one can see the corpuscles pulsating in the tail.

Reproduction

If animals are in the classroom, the **children will see naturally how reproduction takes place**. Rats and mice are easily demonstrated though we have an obligation to find homes for the baby animals. This is a solution to feeding snakes.

Rats are smart and cuddly though they live only 3 years and develop tumors. Birds and snakes will reproduce easily. Metamorphosis can be experienced with dragon flies or **butterflies**.

Dissections

Some people are opposed to dissections for a variety of reasons. Some are political and other are more subjective. But they are important in the classroom. Whenever doing a dissection, **relate the function to the part**. “These are the gills. They help the fish take oxygen from the water”, for example. It's easy to get a fish from the market with all of his parts.

A squid is also a great animal to dissect. Be sure to tell the butcher to leave internal parts in since they typically gut them. **Besides the organs on a squid being very clearly identified, the squid has a rudimentary bone-like thing called a “pen” that looks like a quill pen; it is thin and transparent. It goes throughout the squids body and provides support**. A squid makes ink – which can be put together with the pen and actually written with.

Specimens can also be bought from certain companies; this often includes injections of dye that make seeing the parts much easier.

Be sure that children wash their hands after handling reptiles since they are carriers of salmonella.

Epilogue

Not all this material has to be presented before going on to body functions.

February 22, 2005

Body Function Booklets by Class

Purpose

This material provides a synthesis of the information the child already has from his own experiences and observations and provides new information too. The children learn all animals have hearts but some are 2 chambered, while others are 3 or 4 chambered.

This material takes into consideration **defining characteristics that will classify** animals into the **five classes of vertebrates**. Some of the characteristics are external and some are internal. This material is never presented to the primary child.

The **story material comes before this material** since it provides an understanding for this material. It does this by **relating function to the part** for the first time. For example, the beaver has a big tail that is used for swimming and balance; so there's a **relation between feature and function to point out**.

So when the children come to the Body Function Material, they are familiar with how relations are made.

Materials

For each of the five classes of vertebrates, we find:

- A booklet – each entitled “**Body Functions of _____**” where the blank is mammals or some other group of animals.
- **In the booklet, there is a picture of the characteristic** isolated in red, appearing on the left side of the book; on the right side facing page there is the **definition with the characteristic isolated in red. This booklet is used for control of error.**
- A card with a picture of the characteristics of the group and it is isolated in red. These pictures are the same that appear in the booklet.
- To go with the pictures, there is a set of cards with definitions of the characteristics of the group. The definition isolates the characteristic in red.
- A **set of cards with an incomplete definition**; to complete it, there is a ticket where the characteristic is written in red ink.

(Note there are five classes of vertebra – there are five of each of the items in the list above)

There are **six different body functions**. They are:

- Reproduction
- Respiration
- Skeleton
- Movement
- Protection
- Circulation

At times, a seventh is included and is digestion. But with the purpose of this material being scientific classification, digestion is left out since it does not define the animals in the same manner as the other characteristics.

To accompany this material, one needs the same **picture cards from the Story Materials**, tickets that say the names of the **five classes of vertebrates: fish, amphibians, reptiles, birds and mammals**.

There are also **six cards for the body functions** listed above.

Reading the Booklets by Class Category

Materials

Booklets from the five classes of vertebrates and an assortment of the animal pictures that sample the five classes.

Introduce five names of classes

Begin

(lay out different pictures and name them - "Here you have a trout")

I want you to find all of the animals that protect their bodies with feathers.

(child collects all in that category)

(point out other common characteristics in this group)

I can see animals that protect themselves with feathers fly and many times have feet for perching. We also see they lay eggs.

Animals that have these characteristics, protect with feathers, have wings and lay eggs – we call them birds.

I have a booklet here about birds – so let's read about them.

(teacher and children read through the booklet)

(choose another group)

Let's find out which animals protect themselves with hair?

(categorize this group and again, compare like traits within that group)

Animals with these characteristics we call mammals. I have a booklet here about mammals that we can learn more.

(same process for fish, amphibians, reptiles, birds and mammals)

Epilogue

What we have done is formally given the name of the five classes of vertebrates and their characteristics that define them. Children often know about fish and amphibians, etc. but we are now consolidating that knowledge and formalizing it.

Categorizing and Reading by Definition

Match full definition to card

Materials

Tickets with names of classes on them. Set of cards with complete picture and complete definition.

Begin

(choose any group – we choose mammals today)

(have them lay ticket for mammals on the table)

(have them lay the picture cards across the table horizontally - ticket is column header)

(have them read the definitions and place the card under the picture)



(process for fish, amphibians, reptiles, birds and mammals– eventually mixing and combining before categorizing)

(use the complete booklets as control of error)

Reading by Definition Comparing and Contrasting

Use incomplete definition, comparing and contrasting

Materials

Labels for the five classes of vertebrates: fish, amphibians, reptiles, birds and mammals. Incomplete definitions and their labels.

Begin

(choose a group – we choose mammals)

(lay down heading for the class we have chosen – mammals)

(lay incomplete definition cards down horizontally across table under the heading)

(take name tickets, 'hair' for example, and place above the definition card)

(choose second class, incomplete definition cards – order does not matter)

(read definition tickets and place them next to those definitions)

(point out the similarities and differences between the two classes on the table)

(pick up the tickets, mix them and ask the children to relabel the definitions)

(repeat for other classes – two at a time – contrasting and comparing each time)

Epilogue

Again, the children control their work using the booklets. This is a kind of test for their comprehension of the first two activities.

Correlating Classes and Body Function Fulfillment

Give formal name vertebrate

Materials

Tickets for the names of the five classes. Tickets that have the names of the six body functions. Colored picture cards from each of the five classes of vertebrates.



Begin

(lay names of body function as heading cards on the floor – **last ticket must be 'skeleton.'** - choose any class – reptiles for example)

(make a second heading axis and place the picture cards describing how the body functions are performed – for each given class heading)

(now we have a grid – now systematically point out similarities and differences for each of the body functions)

We have reptiles, birds, fish and amphibians that lay eggs. Mammals do not!

(repeat for other body functions)

They all have a skeleton. They all have a backbone!

This backbone is made up of a vertebra. This is why we call them vertebrates.

(at this point, show again the segmented paper towel dowel to illustrate its movement)

Epilogue

We are forming an **x-axis with class name and a y-axis with body function**. The picture cards describe a particular means; picture of lungs for example – as opposed to gills. This presentation formally presents the name **vertebrates**.

The basic lesson places classes randomly, except the placing skeletons last. A variation is to **list the classes in a specific order**: fish, amphibians, reptiles, birds and lastly mammals. This variation **shows the development in each aspect of body functions** across the five classes. If they don't think of it themselves, bring out the timeline of life and track the development of the vertebrates on the timeline. Here they can see the progression of how body functions have evolved.

October 21, 2004

Introduction to the Classification of Plants

When Is It Given

Prologue

When we take up scientific classification with the children we have to take into account their experiences and what kind of foundation of knowledge they have. One **awareness** the child must have is that of the **great diversity of life** that exists on the earth. The children come to the **knowledge by their own observations** and hands-on activities. The children are aware of the parts of the plants, the parts of those different parts and aware of the parts of animals.

From their explorations the children know that the great variety consists of plants that can have their corollas joined or the petals can be free. Some plants have herbaceous stems and some have woody stems. He will know that some fish have two dorsal fins, some have three and some have none. He will know that some insects have wings and some don't. This **awareness comes to the children through their exploration** because they have the **opportunity to go out**. They can go out to the woods, to the play ground, to botanical gardens and to the library to get books of specimens they may not find in their immediate environment.

Besides an awareness of the diversity of life, the children have also developed their **powers of observation**. As they continue to look for characteristics, their power of observation becomes more refined. They also observe that all living things have needs and that these needs must be met for survival. The child knows that not only does the tortoise need to eat but it eats fruits and vegetables. Children can also see from their observations that how a plant or animal meets its needs help to determine its behavior. It also helps to create the many diversities that exist.

Along side this trained power of observation and an awareness of the varieties, the **child has been given the nomenclature that goes with all of the characteristics**. He knows that joined petals are gamopetalous and free petals are polypetalous. The name attached to a characteristic not only crystalizes that awareness but it allows the child to explore further into the abstract. He now has the tool of language by which he can further his exploration. For example, I know this one is parallel and this one is reticulate, but here's a leaf that appears to be neither – I wonder what it is (the Gingko is dichotomous – the one exception).

Once the children have these basic skills – **the power to observe, the nomenclature that goes with life, and awareness of the characteristics and how they came about** – **then you can present the classification material** without teaching it to them. The child's understanding will be based upon his experiences, rather than from lectures we give. The children will have had some experience with classifying because

they will have done simple classification. But **when they did simple classification, they were only distinguishing between characteristics. When they do scientific classification, they're going to observe the characteristics and then put them into distinguishing groups.**

October 21, 2004

Why Is This Material Given

This material provides the key for a different kind of exploration. Since exploration is a natural tendency for humanity, this material helps the children to meet this tendency. But this exploration provides a new key – an exploration of a different kind. This key is grouping things with like characteristics into one group and differing characteristics into another. This new kind of classification allows the children to order their prior knowledge. And because of this, it helps them to develop the reasoning mind. This is so because the children have to reason out why to put this specimen in this group rather than this other group. This reasoning out and the classifying of information allow the child to build up his intellect. And it also contributes to clarity of thought. It does this by providing a structure around which to order knowledge that the child has already acquired.

Because **this material helps to order the information the child already has, it meets the tendency to order.** Since there is a connection between the natural tendencies and the material, the children find it very satisfying. This material gives the children the opportunity to think and to analyze for themselves whether a plant belongs in this group or that groups – and why. It is through their choice and the process of analysis that the children develop and refine their analytical abilities. These opportunities are given in biology just as they are given in other areas in the classroom - and it is from using the scientific classification material.

October 21, 2004

What Exactly Is This Material

This classification material provides a structure which allows the children to order their knowledge of plants and animals. This structure allows the children to group specimens by identifying and discerning distinguishing characteristics. Once they make a choice, they can place the specimen in the structure in relation to the other groups that make up the structure. This material that we present the children in the Montessori environment is not the latest classification system and it does not answer all of the questions. Because human beings are **driven to precision and exactness**, they are continually searching for further distinctions in classifying things in their world. Consequently, classification systems change continually. So, both you and your children will run across discrepancies in the literature regarding classification systems.

Even botanists cannot agree on which classification system to use. So even today, there is not a universally accepted botanical classification system. And as scientists continue to investigate DNA, new systems are evolving where you classify them based on cellular structure.

When we consider our own classification system, we give the children a basic structure for understanding how classification works. Once they understand this structure, when they run across discrepancies in the literature, their structure won't crumble; they'll merely make adjustments to it if they choose to do so. An analogy can be seen in mathematics: the children learn the decimal system which is a system of base 10. It becomes the structure by which they can investigate other base systems. But without understanding the structure of the decimal system first, it would be hard for them to understand the other systems based on 2, for example. We have to understand what our purpose is for giving this material; we are not trying to make botanists out of the children nor are we trying to create scientists – though this may happen. With this material, **we are trying to give the children the opportunity to analyze, to differentiate and distinguish, to think critically and finally, to decide – based on reasonable information.**

Because those are our goals, the material is designed with limitations inherent. We do not cover all of the possible groups that we could because then there would be nothing for the children to discover. Also, we do not constantly update our material even though it might be old because we want the children to update it when they discover new information. By limiting the material, we create a natural bridge to go into the books. So, the children can thereby learn how to do scientific classification from an actual scientific guide.

The material is designed in a very simple way so that the children can experience all of the groups first-hand. He can see the characteristics and does not have to be told they exist. In the case of DNA, the child would not have the possibility to see for himself. Also this material is given in a dichotomous pattern. This means you have two choices. Is it this or is it that? In this way, our material corresponds to that of the way

scientific guides work. Also, the material is designed for activity. As the children examine the folders – each one individually, they have to put them back, one folder within another folder within another folder. And it is through this activity that they actually learn the taxonomy.

Because we are dealing with both plants and animals. The question arises: “What do I present first? Plants or animals”? In the absence of burning interest from the children, present the plants first. The reason is because plants can easily be explored. It's easy to take a plant apart and look at its ovaries – not so easy with an animal.

The work in the classification material will stretch out over a long period of time. It can begin very quickly after the children have had those experiences with botany in the classroom but once it gets further and further sub-divided, the work slows down. And some of it is more appropriate for the 9 to 12 class.

October 21, 2004

Presentation of Classification

One Way That Classification Can Be Started - 1st Period

Prologue

The first thing one does when presenting the classification material is to take a nature walk. The focus of the nature walk is to **point out everything of diversity** that you find as you go. You want to point out tall plants and tiny plants, plants that live in water and those that live on land, plants that like sun and plants that like shade, plants that make flowers and those that do not.

You also want to point out that **plants survive under all kinds of circumstances**. If you can find a little **plant growing up through a crack in a sidewalk, point that out**. Plants are determined and can survive in dire conditions sometimes.

Then upon return from your nature walk, immediately present the material or you could present it in the afternoon or the next day. But don't wait too long between the nature walk and your first presentation. Gather your children and begin your story.

General Procedure

Take out the heading card. You can read it or the children can read the material. The heading card has general information about the group. When you read the text and there's a picture on the card, relate the text information to the picture. If there are other text cards, read those next. Once all the text cards are read, look at the picture cards and have a discussion about them based on the text cards. Next, take the circle. Each circle will have the defining characteristic of the group on it. **In the case of kingdom Vegetalia there cannot be a defining characteristic because it defines the whole group – this is a difference between the first folder and the ones that follow – testing!**

After you examine the circle, then you go to the subdivision card. It is the one that says what the group breaks into. Read it aloud. Look at the subdivisions and then point to the subdivisions on the cards. Then put your contents away in a precise order:

- circle on top followed by (note this is *presented* just before subdivision card)
- heading card
- any other text cards
- picture cards
- lastly, the subdivision card

Carefully take the contents and place it into the envelope, tie it up carefully and set it on your biology shelf.

Begin

Do you remember how many different plants we saw and how many different kinds of plants there are?

We know that in dry areas, cactus and succulents can grow there. But areas where there is lots and lots of water – plants can grow there too. Water lilies grow in the water. Mango trees like it very swampy. And in areas where it's very very cold, there are pine trees that can live in that climate. Where it's very very warm, we have palm trees.

What about the tallest trees – the Redwoods. We know they are tall, but did you know there are tiny tiny plants that grow in the Tundra. Some plants just live for one season. But some plants live for many hundreds of years. The bristle cone pines have lived for 5000 years.

So, there are all kinds of plants. Because there are so many kinds of plants, they are grouped into a kingdom. And it is called the kingdom Vegetalia. **Vegetalia** comes from the Latin word vegetare - which means to **grow**.

Both plants and animals grow but the animals have movement and they go from place to place. Plants can't move. Once they settle down, they have to stay there. But plants can make their own food. Animals cannot do this. Animals depend on the plants for their food.

I have a folder here and its going to tell us a little more about the kingdom Vegetalia. (the folder Kingdom Vegetalia contains only the circle and text/picture cards at this point)

(read heading card and text cards, then discuss all picture cards with the children. Afterward, take the circle)

This kingdom has been divided into (... see after the circle in Prologue above for remainder of lecture.)

Epilogue

Whenever you present your material, present it with respect and reverence. If you are casual, your children will also be so. This material is hard to acquire, so set the correct example up-front.

Once you present the folder, wait for another day to present subdivisions or depending on the children's interest, you could present the next division. This is guided by the children.

October 21, 2004

Presentation of the Phanerogamae

Prologue

Your folders should be kept in a closet or in the back room – not on a shelf. Go to your cupboard and pull out your folder and say: **Note:** Each kingdom, branch, etc. has “contents” and a “folder.”

Begin

I have a folder of a group that is called division Phanerogamae. The word **Phanerogamae comes from two Greek words, phaneros, which means visible, and gamae, which means marriage.** These plants have a visible marriage.

Why? Because we can see flowers.

Look at the picture on the front of the folder. You can see that this is a visible marriage. Can you think of other plants that have a visible marriage? (name any flowering plant, oaks, cactus or pines)

Let's see what's inside!

(take out contents. Read heading card, read other text cards, look at pictures and relate them to the text just read and then discuss picture cards while looking at them and discussing attributes highlighted in the text. Then look at the circle with the identifying characteristic. In this case – visible marriage (a flowering plant))

(open inside folder, point out defining characteristic on the circle is the same as the defining characteristic on the folder. There is a list of characteristics on the inside of the folder; read them to the children. “... they reproduce by means of seeds”)

(note the subdivision groups are also listed inside the folder – this helps put the material away correctly and should not be read)

(Once characteristics are read, go to the subdivision card. This is where they get the subdivisions “Division Phanerogamae divides into two” Point out the pictures of the two subdivisions)

(Once you finish reading the subdivision care put them away

- circle
- heading card
- other text cards
- picture cards
- division card)

(Put the contents into the folder and tie it up for safe-keeping. Place the Kingdom Vegetalia folder on the shelf since it is now being used. Each new folder brought out subsequently, is taken from the closet, since it was heretofore not being used.)

Epilogue

Present the contrasting folder next unless interest takes you elsewhere – that is, into subdivisions of Phanerogamae. Present Cryptogamae exactly the same as above

October 21, 2004

Presentation of the Cryptogamae

Prologue

This word comes from two **Greek words, cryptos – which means secret.**

That is, there was a secret marriage. Scientists gave them this name because they didn't know how they reproduced.

Begin

Present this whole subdivision as Phanerogamae was presented.

Epilogue

Give experience with Cryptogamae before going into this presentation.
Ferns in terrarium. Grow horsetails in the classroom. Rubbings of ferns.

To put this folder away, open up kingdom Vegetalia and put it with division Phanerogamae This method of organizing the material presents the concrete organization of the abstract divisional structure.

October 21, 2004

Branch Angiospermae

Prologue

Start with Angiospermae because children are more attracted to them and they are more common.

Begin

The plants in here are from the branch **Angiospermae**. This word comes from two Greek words, aggeion – which means **receptacle**, and spermae – which means **seed**.

They have seeds that are in a **closed receptacle**.

(if children have cut open ovaries, this name will make sense)

(open folder, read contents as with previous, replace into Angiospermae within Phanerogamae folder, within kingdom Vegetalia)

Branch Gymnospermae

Same lesson outline as above; striking difference is there is **no receptacle**. Gymno means **naked** and spermae means **seed**. These naked seeds are found in cones (which are not flowers, hence, no receptacle).

Epilogue

Take the children out to a place they have already visited, point to a plant and ask them, “Is this from the division Phanerogamae or Cryptogamae. They will answer and you will ask them why? Give them an opportunity to use their new terms and to analyze.

Pattern of the Presentation of Each Envelope

Note: presentation of the classification material is like a 3-period lesson. We have presented the 1st period. The 2nd period comes in tracing the genealogy of a given specimen, in which we still present names.

- Select a folder. It can be either a new subdivision or a contrasting group.
- Give the name of the group and the etymology of the name.
- Point to the picture on the front of the folder, pointing out the characteristics.
- Remove the contents from the folder.
- First read the heading card, then the remaining text cards, look at picture cards, examine the circle that has identifying characteristics,
- Examine the inside folder, relate the circle to the picture on the inside folder, read the list of characteristics
- Read the subdivision card
- Show how to put the contents back in a precise order from top to bottom: circle, heading card, text cards, picture cards and subdivision card.
- Place contents back into the envelope, tie it up and put envelope away.

Epilogue

You can see how the material is dichotomous. It's either this subdivision or that subdivision. You can see the activity involved, in putting one envelope in relation to another envelope. It is through this activity of putting one away, within another, that the children will learn the subdivisions.

You can see that the characteristics are ones which the children can actually experience and see with their own eyes.

Then, there is the etymology. Within each name, the characteristic is hidden. The etymology tells what the characteristic is.

October 21, 2004

Another Way to Present Classification

Prologue

If there is no preference, present Dicotyledonae first because most plants in the world are in this Class.

Begin

Take the folder from the cupboard, take it out, read the contents and then put it away:

Open up kingdom Vegetalia, choose Phanerogamae or Cryptogamae (choose Phanerogamae), choose Angiospermae or Gymnospermae (we choose Angiospermae), put the rest away, keeping contents on the top.

Present this Class (Dicotyledonae), followed by the contrasting Class (Monocotyledonae).

Epilogue

You can do up through this point quickly in the 6 to 9 years old class. After they go into books, they have little use for these materials. Break it down and put it away for the year.

October 21, 2004

Dicotyledonae

Prologue

The folders will be taken out, investigated, and put back, in this presentation. We begin at the Class level for this. The children already have experience with Monocotyledons and Dicotyledons; this is why we begin at the Class level – and then we work upward.

Have real specimens on-hand and start with Dicotyledons. On the real specimen, examine the physical characteristics.

Begin

This plant has its flower parts in groups of fours or five. It has leaves with reticulate veins – and it probably has a tap root.

Flowers that have these characteristics belong to a group called Class **Dicotyledonae**.

The word Dicotyledonae, comes from Greek – di for **two** and cotyledon, which means **cup-shaped cavity**.

(present the folder as we did in the first method of presentation of classification)

(tie up the folder, as you put it away, tell the children...)

This group of plants belong to another group. It belongs to the branch Angiospermae.
(say no more, set it on the shelf)

October 21, 2004

Monocotyledonae

Prologue

Have a real specimen whenever possible.

Begin

Look at this flower.

Its parts are in groups of threes. The roots are fasciculate. The vein pattern of the leaf is parallel.

Plants with these characteristics, we call Class **Monocotyledon**, coming from the Greek words mono – meaning **one** and cotyledon – meaning **cup-shaped cavity**.

(present the folder in the same way as previous lessons)

The Class Monocotyledonae, just like the Class Dicotyledonae, belongs to the Branch Angiospermae.

(place it on the shelf next to the folder for Dicotyledonae)



Next time say: “**Both the Monocotyledons and the Dicotyledons have seeds that are in a closed receptacle.** So, they belong to a group called **Angiospermae**. I have a folder on the Angiospermae

(Give the etymology and give them the folder for Angiospermae. When finished with the folder, take it to the shelf and put Monocotyledons and Dicotyledons into it. Tie it up)

The Branch Angiospermae belongs to a group called Division **Phanerogamae** – as does

Branch Gymnospermae.

But **Gymnospermae have an open receptacle unlike the Angiospermae**. Gymno is Greek for **naked** and spermae means **seed**.

(Introduce them to the folder.)

Gymnospermae also belongs to a group called Division Phanerogamae.

(set the Gymnospermae next to the Angiospermae) (Note: Always give etymology)



Branch Angiospermae and branch Gymnospermae both belong to a Division Phanerogamae.

The Division Phanerogamae belong to the kingdom Vegetalia.



Present the contrasting folder Cryptogamae. Cryptogamae comes from Greek, crypto meaning secret and gamae, meaning marriage.

(tie it up and set it on the shelf, mention it also belongs to the Division Phanerogamae.



(bring out the Division Cryptogamae – do the normal presentation – etymology, etc.)

Division Phanerogamae and Division Cryptogamae both belong to the Kingdom Vegetalia.

October 21, 2004

Continuation of the First Way We Began

Begin

Today, we need to find what are the subdivisions for the Dicotyledons. Let's choose the Subclass **Archichlamydae**.

Archichlamydae comes from two Greek words, archi, meaning **ancient** and mydae meaning **dress**.

These plants are Polypetalous.

(Go through the folder in the same way as in previous lessons, i.e., heading text card, other text cards if present, picture cards for discussion, circle with characteristics).

Epilogue

Afterward, take folder Archichlamydae, place it in the Dicotyledons folder, which is placed in the folder for Class Angiospermae, place that in the folder for Division Phanerogamae, which is in turn placed in the folder for Kingdom Vegetalia.



Present contrasting group **Metachlamydae**, Greek for Meta – meaning **recent** and mydae – meaning **dress**.

Plants in this group have flowers that are gamopetalous.



Begin

Let's talk today about the **subdivisions of the Class Monocotyledonae**.

Here we have one of the subdivisions of the monocotyledons. This group is referred to as Subclass Group I . (take out folder material and examine as customary)

All plants that have flowers that belong to **Group I do not have floral envelopes or otherwise called a perianth. A perianth has a calyx and corolla.**

Epilogue

They include grasses, sedge, cattail and arum. All the contents are put away hierarchically as usual.



Group II are monocotyledons that have floral envelopes.

Epilogue

Go on another nature walk. The children have experience under their belts. They have lots of experience putting folders one into another which gives them the impression on how one group is related to another.

When you take them out, ask them questions. “What Class do you think this plant belongs to.” Examples are Iris, Lilies, Orchids, Spider Plants.

October 21, 2004

Families

Before coming to the family level, you should have presented everything that is in the folder now. This means you will present the families at the 9 to 12 year old level most likely.

At the family level, our materials are no longer dichotomous. There are many many families. We have selected only a few to go into here. One of the difficulties with Families is that the characteristics are hard to see. We have picked the simplest ones.

When you come to the family level, you should have a whole specimen – so you can look right at the characteristics. It is always better to get a wild specimen rather than one that is bought at a nursery because store-bought plants have been tampered with and their characteristics may have been changed.

Because you want the children to discover certain families, it is important to prepare the environment – so you can direct and guide their explorations and their discoveries. You want to have in the environment a few kinds of plants from the same family. When they discover these, they will see that two different kinds of plants have the same characteristics. This allows them to have some repetition for the characteristics of a particular family.

It's also nice to have some plants in your garden that do not appear in the Families found in the material because then, the children will have reasons to look into the books and to classify that unknown plant into its Family.

Because the material provides you with many families, you have to decide which one to give first. The answer is to give the family called Cruciferae. (We don't have any so will present Roses)

Cruciferae is an easy family – radishes, Brussels Sprouts and mustard. Cruci means cross.

Begin

(have children examine the characteristics of your specimen)

There are many stamens that are turned toward the pistils in this specimen. I have a folder here that will tell us more about these characteristics. They belong to the family Rosacea.

(at the family level, bring out the circle and look at the characteristics on the circle and compare them to your real specimen. on the circle, one sees the stipule, for example, and we can see this in the specimen. Read the heading card and look at the pictures as usual)

Epilogue

In the folders at the family levels, there are fewer cards – less contents, since we are so refined in our classification. Also note the inside of the folder; every other folder, up to this point, had the groups into which this groups subdivides. We do not have that since we are at the bottom level of the materials.

Families of Subclass Archichlamydae include Ranunculaceae, Rosaceae, Cruciferae and Leguminosae. All these flowers have ancient dress which means polypetalous.

Subclass Metachlamydae includes families Compositae, Scrophulariaceae, Polemoniaceae and Labiatae. Subclass Group I families are Graminae, Typhaceae, Cyperaceae and Araceae.

Subclass Group II includes Amarylidaceae, Liliaceae, Iridaceae and Orchidaceae.

A great book is called “Plant Families” by Carol Lerner. It has illustrations on one side and information about them on the other.

October 21, 2004

Book Work

Prologue

There should be an actual classification guide in the classroom. We can find guides that classify in ways that are not specific. Don't buy those. Where ever we teach, go the local university and buy their scientific guide. It does not matter how thick it is – that it is a tome. It is easy to use.

Scientific guides have physical characteristics and generally they are the physical characteristics that can be seen and experienced by the children. They are also dichotomous – meaning the user is given a choice between this or that. If this, go to page 'this-page', else go to 'that-page.'

When you first introduce the guide to the children, you will rely on their exploration. If you come across a term that you don't know, check the glossary. There is a glossary in the back of every good guide. Look up the term and carry on. In scientific guides, they will give you many characteristics to look at. But if you can determine what you are looking at by looking at the first characteristic, you don't have to read through all the rest; make your choice and move on.

So, use the guide to find the family. Once you have the family, you want to find the name. Then you can go back and use the circles from the materials in order to examine and match characteristics.

October 21, 2004

The Genera Level

We are coming down below Families. At this level, the children have to be able to do book work because your material does not go down to this level. You need to have two specimens from the same family because now you're going to distinguish and subdivide that Family.

The first thing you do is point out the similarities and the differences between the two specimens. Similarities might be "Leaves are narrow and linear. Stamens are three in number and the ovaries are inferior." The differences might be described in this manner: "These flowers are arranged on a spike inflorescence (arrangement) in the nomenclature material."

For those who are really interested, introduce them to that material. Show the Spike inflorescence. There is terminal inflorescence and spike inflorescence. (For example, Gladiola = spike = Corm. Iris = terminal = Rhizome.)

At this point go to your book and you can access the genera level information. Once the children do this, provide the children with the circles of the materials; they can make circles for each of the two specimens using the characteristics that distinguish them, taken from the guides. They can also make a folder if they like or they can make additional circles. These circles can be added to the classroom nomenclature material.

For the exam, know the names of flowers and ovaries, the names on the folders and the basic subdivisions, the etymology and how to use the material.

October 22, 2004

Tracing the Genealogy of a Plant - 2nd Period

Prologue

Children must have experienced everything in the folders before this presentation can happen. Therefore, it will probably happen in the 9 to 12 class. The children in the elementary class like to work in groups and this is a group activity. There will be discussion based on making a choice. When we present this, we should have a real specimen. We also need to have the folders all packaged up and present.

Begin

Does this specimen belong to the kingdom Animalia or the kingdom Vegetalia? Why?

Have them go to the material and get the circle for the kingdom Vegetalia. Place the circle on the table.

Does this belong to the Division Phanerogamae or Cryptogamae? Phanerogamae. Yes, **Phanerogamae**. Why? It's got a **visible marriage since it has flowers**.

(Get the circle for Phanerogamae. Superimpose the circle for Phanerogamae upon the circle for kingdom Vegetalia.)

Is this specimen from the Branch Angiospermae or Gymnospermae? **Angiospermae**. Why? There are **seeds in a receptacle**.

(Get the circle and superimpose it onto the other circles above.)

Does it belong to Dicotyledonae or monocotyledon? **Monocotyledonae**. There are **parallel veins and flower parts are in threes**. Get circle for Monocotyledon and take the circle.

Superimpose the circle upon all of the others.

Does this specimen belong to Group I or Group II? Yes, it is Group II. Why? We have to memorize this – it has a perianth.

(Take out the circle and superimpose it onto the others.)

Does this specimen belong Iridaceae, Orchidaceae, Liliaceae or Amarylidaceae?
Iridaceae? Why? Because it ... (fill in the reason based on what the specimen has. Take the circle for the family level and superimpose it.)

Now we have all the circles superimposed on each other.

Read to the children the genealogy of the specimen. "This specimen belongs to the kingdom Vegetalia, Division Phanerogamae, Branch Angiosperm, Class Monocotyledon, Subclass Group II, Family Iridaceae."

Let's do another one!

Epilogue

If you don't put it all back together each time you present a specimen, things will be misplaced. This is a second period because I gave them the names and asked them to identify which it was.

October 22, 2004

Tree of Classification - 3rd Period

Material

All of the circles from all of the folders, grouped according to size. Branches that correspond to each of the Subdivisions, also grouped according to size. A folder in which you store your branches labeled “Branches for Tree of Classification for Kingdom Vegetalia.”



Prologue

This is the 3rd of the three-period lesson. The 1st period is the presentation of the envelopes. The 2nd period is the tracing of genealogy for a given specimen. Note there is no logical order to the questioning below.

Begin

(Present them with the circle for Kingdom Vegetalia. Place it on the floor. Since they have worked on this for awhile, they will know these answers.)

In our material, **how many Divisions are there in the Kingdom Vegetalia? What are they?** Yes, Phanerogamae and Cryptogamae.

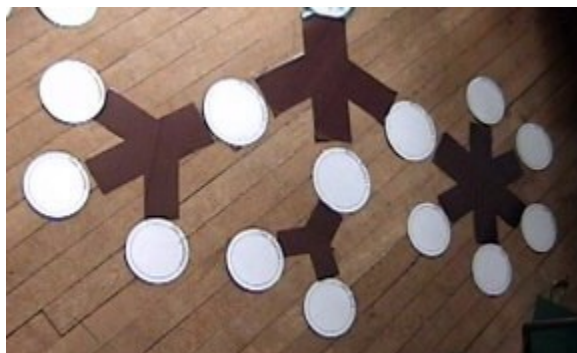
Kingdom Vegetalia, and start of Divisions Phanerogamae and Cryptogamae



(Place branch on a hierarchy following kingdom as in a tree. Place Phanerogamae and Cryptogamae at end of the split branches.)

It does not matter which branch we take to complete the tree at the next level. Let's do Cryptogamae.

How many subdivisions does the Branch Cryptogamae have? Yes, three. What are they? Pteridophyta (Ferns), Thallophyta, and Branch Bryophyta (mosses). Place 3-branch attachment to the Cryptogamae stub and place subdivision circles at the end of each of the 3 that branch off.



Division Cryptogamae

How many subdivisions do we have for Phanerogamae. Two. What are they? Angiospermae and Gymnospermae. Place new branches and circles downward in hierarchy. We take any branch we like next, continuing down the hierarchy.



Division Phanerogamae

How many division in Angiospermae. Yes, two. What are they? Monocotyledon and Dicotyledonae. (Place branch and circle.)

How many divisions for Monocotyledon? Two. What are they? Yes, Group I and Group II (place tree branch and circle)

How many subdivisions do Gymnosperm have? Three. What are they? Coniferales, Cycadales and Ginkgoaceae. (Place branch and circles.)

How many subdivisions does Bryophyta have? Two. What are they? Hepatica and Muski. (Place branch and circles.)

How many subdivisions does Thallophyta have? Yes, Five. What are they? Phaeophyceae, Flagellata, Diatomaceae, Eumycetes and Lichens. (Place branch and circles.)

How many subdivisions are there for Dicotyledons? Yes, two. What are they? Archichlamydae and Metachlamydae. (Place branches and circles.)

How many subdivisions are there for Metachlamydae? Yes, four. What are they? Labiatae, Scrophulariaceae, Compositae, and Polemoniaceae. (Place branches and circles.)

How many subdivisions are there in Archichlamydae? Yes, four. What are they? Ranunculaceae, Rosaceae, Cruciferae and Leguminosae. (Place branches and circles.)

How many subdivisions in Group I? Yes, four. What are they? Graminae, Typhaceae, Cyperaceae, Araceae. (Place branches and circles.)

How many subdivisions in Group II. Yes, four. What are they? Amarylidaceae, Liliaceae, Iridaceae, and Orchidaceae. (Place branches and circles).

Epilogue

When you do this activity, it's when the child gets the whole impression for the classification system of the plants. Sometimes, because of the limitation of material, they can see a gap and that's when they want to develop the material. It would be fine for you to have these circles that are blank because that calls for the children to find the characteristics. If you have a good botany book or good classification guide, you will easily be able to find the characteristics.

Sometimes, when they look at the material, and because they have been making their own investigations, probably looking into books and visiting botanical gardens and universities and talking to experts about the plants, they may discover some of the discrepancies in our classification system. For example, Branch Thallophyta were at one time classified as plants but now biologists have broken some of them into their own kingdoms. This is a point of discovery for the children. We could change the material as teachers to reflect this; but it would take the excitement out of the discovery for the children as well as their opportunity to do a Great Work. They also delight in finding the teacher wrong – and you will say - maybe you want to do something about it. (We have a special branch for this).

The whole classification scheme is presented at this point to the children in its full impression. Remember, their enthusiasm depends on your own.

October 22, 2004

Description of Material

Prologue

As you already know, every group is placed into a folder of diminishing size, going from largest to smallest, from kingdom to the families.

Folders

The colors of the folders vary in degrees in shades of green, going from darker to lighter, from the kingdom to the families.

Each envelope on the outside has the name of the division as well as the picture that represents the group as a whole. The only exception to this is the kingdom folder which only has a name.

On the inside of all of the folders, except for the kingdom folder, you have a picture of the identifying characteristics and that picture is the same as the picture on the circle that goes with the folder.

Also inside the folder, is the list of characteristics that define the group.

On all of the folders, on the inside flap, except for family folders, you have the subdivisions listed. Those divisions are listed there in order to help you and the children put the material away properly. They are not used for purposes of presentation.

Contents

In each folder you will have a circle. That circle has on it a picture of the identifying characteristics of the group. It also has the name of the group and the subdivisions to which it belongs. The name border of the circle is about $\frac{1}{2}$ inch wide around the circumference.

There is a heading card. This names the group with general information about the group. There can appear a few pictures on the heading card that relate to the information in the text. Depending on the degree of subdivision, you may have other text cards. On these text cards, there also can be small pictures that relate to the information in the text.

At the family level, the cards are smaller and so there are no pictures on the text cards. Also, there are fewer cards at the family level since we only need the distinguishing characteristics of that family.

There are picture cards. The pictures on the cards relate to the information derived from the text. The last card is a subdivision card. It names the groups that the

group itself will subdivide into.

The contents of each folder will be mounted on the same color as the folders in which they belong.

The contents are always put back into the folders in a precise order and on top of any other folders.

Branches

We have a main branch (brown for the hierarchical tree) which narrows by ½ inch as it descends to subdivisions. The main branch of the next subdivision corresponds in size to the preceding subdivisional branch. Its subdivisions are ½ inch more narrow than its entry or higher main stem (branch). And so on. **Trace these before you leave the institute – they are difficult to make!** (Make them out of a firm material paper.)

Pictures

Any of the circle pictures and written text are already provided. **You have to collect all of your pictures.** Get used books and cut them up. “The ABCs of Nature” is a great book for that purpose. You can also get it online. You can also get them color printed when you find a specimen you need. “Ranger Rick” magazine, National Geographic and so on are good. Catalogs are also useful – especially “White Flower Farm” (found at the address Plantsmen in Litchfield, CN 06759). This is a good catalog because they include Dicotyledons.

February 23, 2005

Introduction to the Classification of Animals

By the time you get to animal classification, the children have built up awareness of them. The experiences they have had are that they've cared for the animals in the classroom, they've observed them and observed animals outdoors as well. We hope they have had many opportunities for going out. They may have gone to animal reserves, the zoo, the forest, tide-pools, ponds or their own back yards.

The children have also had the opportunity to work with the material in the classroom – beginning with the **Story Material which we hope launched them into their own investigations about animals.** We hope they explored good books about animals. And then we hope they've had more experience with the particulars of animals through the **Preparatory and Parallel for the Body Function Material activities.** **These activities were on-going during the children's use of the material and helped provide understanding about characteristics that lead to scientific classification.** We also hope the child has had a chance to dissect animals so they can see for themselves the internal structures of the different kinds of animals.

The children worked with the **Body Function Material and they were given the names of the five classes of vertebrates and became aware of the characteristics that defined each of the groups.** They also had the opportunity to do simple classification – a small but important activity that allowed them to sort and organize the new information they have discovered.

As the children were working with, observing and exploring animals, they became aware of two important aspects. The first aspect is they discovered the **features developed within an animal type is related to meeting its needs for survival.** Consequently, there is a diversity of life on earth. They also became aware that **animals have different behaviors based on meeting their needs for survival.** So when the children come to the classification material, they have a sound basis and a body of knowledge for understanding it.

Material Description

All of the principles pertaining the plant classification pertain to animal classification as well. This means there are **folders that vary in size** going from larger to smaller as you work from Kingdom down to Family.

Within each of the folders, one finds the contents that are:

- a circle with defining characteristics of the group
- a heading card that has general information about the group

- text cards that provide details about the group.
- picture cards that illustrate the points expressed in the text

The last card is the subdivision card. Just like with plants, the Kingdom folder does not have a picture since it is the top level with no single defining characteristic. At the next level down, on the outside of the folders, there is the name of the group and a picture of a representative organism.

Inside the folder at this second level, one finds the picture of the characteristics and it is the same as the circle in the folder. One also finds a list of the defining characteristics and a **list of the subdivisions – purpose of which is to help put the materials away**. The envelope should vary in color going from darker to lighter – progressing from Kingdom to Family. But it is difficult to find enough shades of red generally. So, the placard on which the name appears and the rim of the circle can be colored as opposed to the entire folder and this serves the same purpose.

Both plants and animals have Kingdom. Plants have division whereas animals have Phylum. Plants go to branch and animals go to Subphylum. Plants and animals both go to Class. The plants go to sub-class whereas animals go to Order. Then both go to Family, Genus and species.

The classification material is limited since we want the children to develop it. However, one needs guidance when the children want to develop a group. So, for the Phyla that we choose, there are **points about the distinguishing characteristics one should look for**. These points can be kept in mind whether used for making of material or telling stories. There are so many Phyla that we must limit ourselves to a certain few.

The **Phyla we pick to make the children aware of** are:

- those the child can have first-hand experience of
- those they can actually see whether with the eye or microscope
- those that are available and tangible

Phyla that are most interesting and accessible include:

Protozoa	(G. first animal)	amoebas and flagellates
Porifera	(L. pore bearer)	sponges (common for bathing, glass and a third)
Nidaria	(G. Hollow gut)	jelly fish, hydra
Annelida	(L. little rings)	earth worms, leeches
Mollusca	(G. soft body)	oysters, clams, snails, octopus, squid
Echinodermata	(G. hedgehog skin)	sea lilies, sea cucumbers, starfish
Arthropoda	(G. jointed foot)	centipede shrimp, spiders, insects

Chordata (G. string) contain the five classes of vertebrates

The **aspects draw attention** to are:

- basic anatomy
 - symmetry in the body design (single-cell [none], radial and bilateral for worms and others)
 - segmentation (body broken into parts – more advanced than symmetrical bodies)
 - cell layers (two or three layers – ectoderm (skin), endoderm (gut cavity), mesoderm (organs))
- movement (pseudo-pods, cilia, tube-feet, limbs, wings)
- skeletal support (spines, silicon, shells, pen, exoskeleton)
- reproduction (asexual [budding, fission, conjugation], sexual (monoecious [hermaphroditic like snails] and dioecious [requiring one of each sex to combine])
- nervous system (none, ganglia, brain)
- circulation (hearts or not, blood or not, veins or not)
- respiration (water absorption, air absorption)
- digestion (how and what is eaten)

The materials for animals is limited partly because children are interested in animals and partly because we want them explore using first-hand experience, books and projects.

Sometimes we need to encourage children to move beyond their favorite animals. Be a story-teller about animals. Create them about the different Phyla to bring awareness. Bring the salient points about each group our in the stories.

February 23, 2005

Presentations

Envelopes

(first period – done exactly as with plant material)

Take folder from shelf. Introduce the group by giving the name and etymology of the name. Take out contents of the folder, read heading card, read text card, look at picture card, look at circle with defining characteristics of the group.

Open inner folder and compare circle and inside of folder, focusing on group characteristics. Read characteristics on inside of folder. Examine the subdivision card including the subdivision names pictures of animals in the subdivision. Put contents of folder away in same order as done with the plant material. If envelopes in the folder, place contents in top of the folders. Put folder on the shelf in relation to any other folders already presented.

Genealogy

(this is the second period lesson)

When doing genealogy, have a live specimen. First question is, “Is this from the animal kingdom or the plant kingdom? Why?” (Animals can move about but cannot make their own food.) “Get me the Kingdom Animalia circle.” They place the circle on the table.

Next question is guided by the extent to which material has been developed. If there is two Phyla, “Does this animal belong to the Phyla Cordata or the Phyla Arthropoda? Why?” Because it has a cord that goes through its body. Ask them to get the circle from the Phylum Chordata. Have them super-impose it onto the Kingdom circle.

Next question is also geared to the material; “Does this animal have a vertebra or not? Yes, it does. So what group is it from?” Sub-phylum Vertebrata. Get the circle and super-impose it onto the first two circles.

“Does this animal belong to Class Mammalia, Aves, Reptilia, Amphibia or Pisces? (amphibians) Why? It protects its body with moist skin. Have the child take the circle from class Amphibia and super-impose onto the first three circles.

Then ask the name of the three orders from which the animal comes. “Does this animal come from Anura (w/o tail), Apoda (w/o legs) or Urodela (visible tail)?” Anura. Why? Because it has no tail but has legs. Have the child get the circle for Order

Anura and super-impose it onto the group.

Next we ask about families. At the family level, one can make use of the circles when selecting from which family it comes. Is it from the family Ranidae, Pelobatidae, Hylidae or Bufonidae? Ranidae. Why? Because the animal is thin. Super-impose the last circle onto the group of circles.

After having gone through all the material, then read the genealogy for this animal off the rims of the circles. Announce the names of the groups from Kingdom down to Family along with the characteristics.

Tree of Classification for Animals

(this is the third period lessons)

Take all circles out of the folders and group them to size. There are also branches which are grouped to size. Place circle for Kingdom Animalia on the floor and ask, "How many subdivisions does this have"? This is a tricky question, since we don't give all the Phyla – but we have made two branches in the materials. At the institute, there is a third teacher-made Phylum. Begin by laying out the branches on the floor.

Have them get circle for the Arthropods. Protozoa, Chordata, Porifera, Coelenterata, Annelida, Mollusca, Echinodermata follow. Lay down circle for each.

Which Phyla should we explore first? Arthropoda. How many subdivisions? One. What is it? Subphylum Chelicerata. Place it on the branch. Continue downward: Arachnida, Crustacea, Myriapoda and Insecta. Place circle for each at end of branches.

Next let's do Phylum Chordata. Sub-Phyla Vertebrata. And it includes Classes Pisces, Amphibia, Reptilia, Aves and Mammalia. Class Amphibia has Orders Urodela, Apoda and Anura. Anura includes Families Ranidae, Pelobatidae, Hylidae and Bufonidae.

(repeat per pattern down through the subdivisions)

Next let's do Phylum . And it includes Classes. And this includes Orders

There is no classification guide for Kingdom Animalia since it is so huge. There is classification guides for insets, for mammals perhaps – but never for the entire Kingdom. Take guidance from the back of a good used college zoology textbook. See how divisions and subdivisions are handled in that text.

There are many parallels between Kingdom Animalia and Kingdom Vegetalia. One is that the children have **lots of experience at the Class level with animals. Which piece do they get it from? The Body Function Material.** So when presenting envelopes, just like with plants, enter at the Class level and work upward to the Kingdom level.

In anticipation of **exams, keep in mind similarities between the two classification systems**. The slip says scientific classification, whether plants or animals – have similarities and differences between them: both begin with Kingdom working downward – with similar names. There is limitation of materials, choosing subjects they will have had experience with.

February 23, 2005

Ecology

The subject of ecology has been recognized since the time of Aristotle. It became an organized science in the 18th and early 19th centuries. The word ecology comes from the Greek word '**oikos**', which means **house** and **logos**, which means study or **reason**.

Ecology is the science of relationships between organisms and their environments. The Greek derivation makes sense – the study of the house. The **essential understanding behind ecology is that of interrelatedness and interdependence**. Organisms interrelate with each other and are interdependent; this all depends on the environment in which they exist.

Another thing to become aware of, when we think of organism and environment, is that there is a delicate balance between the two. In modern times, human beings have the power to have an inordinate affect on the environment – favorably and non-favorably. Due to this, we have to understand how important it is to maintain that delicate balance.

Rachel Carson said, “To the natural forces that act upon wild populations of plants and animals, the influence of humans has been added in many places and with increasing vigor in recent centuries and decades.” Maria Montessori also thought it was important that the children be aware of the factors of life and earth, and their interrelatedness. As the children study, first the timeline of life and then the coming of human beings on two timelines, the children study these factors of life in relation to the physical environment.

Because all aspects of geography, history and biology are treated with the principles of Cosmic Education, it soon becomes apparent to the children that the elements of life and their environments are interrelated and interdependent.

Geography

When we looked at the physical features of geography, we considered the work of the water, the work of air and the work of the sun. The child comes to see how these three factors working together change the face of the earth and then after time, changes it again.

The children realize that this is a work that never completes; it is on-going and it continues today.

Biology

When children investigated plants and animals, we drew their attention to the conditions of the environment in which each of them lived. We never spoke of plants and animals in isolation – but always in relation to their environment. For example, we talked about the fish's body being shaped in such a way that it could move through the water easily. We talked about when roots sought water underground, they grew into a network of roots that helped prevent erosion.

When we talked of leaves of plants, we noticed the variations were dependent of where the plant lived in the world.

History

When we spoke of people of the past, we talked about what their lives were like in relation to their physical environment. With the Question and Answer History chart, we spoke about the flora and fauna of the place and time where each group lived. We did this not just to create the past vividly, but to highlight the factors of the environment that influence the lives of people at a given time and a given place. For example, from the mammoths, people were able to derive a supply of food and materials. From the papyrus growing on the bank of the Nile River, the Egyptians were able to make paper.

Cosmic Tasks

Dr. Montessori was so aware of the relationship between life and environment that she spoke of **cosmic tasks** – as though each element in nature, by merely existing, performed a task maintaining the balance that has evolved throughout the ages. **It is as if everything had a double-purpose in existence.**

One purpose is **immediate** – **and it related to survival** of the particular plant or animal. The other purpose, which is remote and unknown to the species surviving, is related to the creation and maintenance of the balance of nature. Dr. Montessori gave the **example of the corals.**

The coral's first task is to stay alive by drinking gallons and gallons of water. At the same time, they are performing the second task – which is the purification of the environment, the water. Since, as they drink those gallons of water, **they remove calcium-carbonate from the water – a substance that is toxic to other animals.** But they **used the calcium-carbonate to build their shells.**

This **double-purpose** can be seen again and again with the animals. The earthworm's preferred diet is soil. And as it eats soil, it aerates the soil. This makes the

soil a better climate for plants to live in. The bee has an attraction for nectar and as it searches for its nectar, it helps support pollination. Micro-organisms feed upon putrefied matter. While they do this, they rid the world of waste. They, in turn, convert it to carbon and nitrogen – elements that other organisms depend upon.

If we look at plants, we see they too perform cosmic tasks. Just by living, the plants rid the atmosphere of carbon-dioxide. At the same time, they give off oxygen. Plants also help to prevent erosion and they help to rid the soil of extra water. Specific examples are kudzu – brought from Japan to the south of the United States – but now it's now taking over!

The **reeds that live in Holland help to reclaim the land from water by sucking up the water**. Pine trees take acid out of the soil, making it conducive for growth of other plants.

When we think of cosmic tasks, we can also think of **inanimate tasks** like air and water. They seem to perform tasks just by living by their own natures or laws. For example, air cannot help but go up when it is heated. Hot air cannot help but carry water. Because the two laws are followed, they collude to bring rain. When rain is generated, it waters the plant life in the environment.

When Dr. Montessori thought about the animate and inanimate things, she also wondered about people – thinking they must be part of this plan. Perhaps people were working for their own selfish needs for survival but at the same time, perhaps they were also carrying out a cosmic task – unbeknownst to them. But because humans possess the ability to see themselves as humanity, outside of themselves, perhaps they are in a better position to discover their part in the cosmic plan. By making this discovery, they could perhaps fulfill their plan with conscious collaboration.

Viewed in this way, the **whole world becomes one ecosystem**. It is a closely knit unit which includes the environment and all living beings in it, including humanity. With the children, at some point after they've investigated the world of biology, the stories of history, the lessons of geography – then make them **aware of the interdependency** between the various elements of nature. This will **usually happen at the latter part of the 6-9 class**. Don't give it too soon because it is such an important story that we want the child to understand the importance of it. Then they too can begin to consider their part in the cosmic plan, and how they might make a contribution.

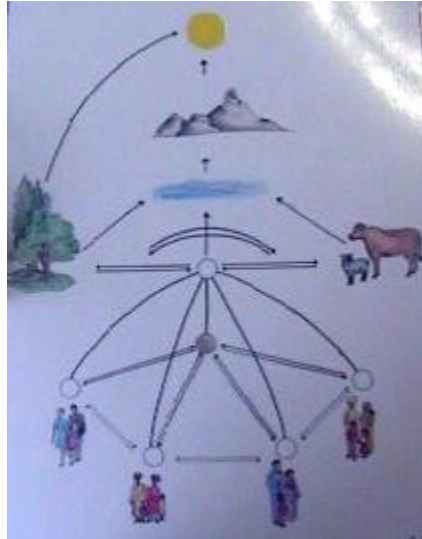
Presentation

Prologue

Tell them a story. Gather a small group around and point to the sun.

Material

Chart of interdependence (chart S)



Begin

Here we have the sun (point)

Here are the rocks or land (point below to the rocks)

The sun heats the rocks. When it heats the rocks, the rocks crumble and then they can support life.

But the rocks need the sun for this. So we can say that the land depends on the sun.

Here's the water. (point central area with water)

The water depends on the land because the water needs the hot land to start the water cycle.

The land is heated up before the water. This creates an updraft of air that carries the water onto land. In this way, the earth can be covered with vegetation.

Without the land, the water cycle would not be possible. So, we say the water depends on the land.

Here we have the plants (point to plants on chart)

The plants depend on the sun for warmth and light they need to make their food.

They also depend on the water, since plants need water to live.

The plants also depend on the animals because they need carbon-dioxide animals provide to create their food.

The animals also depend on water and they also depend on plants for their food.

Here we have people (point to people on chart)

People have spread to the four corners of the earth. We can say that all of the people make up humanity. This circle (center of chart) here, represents humanity.

We know that people need water. People need plants for oxygen and food. We also need other animals because we use them for food and for material and labor.

What is interesting, is that we can say that plants and animals depend on people too, because people have been agents of change and propagation for both plants and animals.

Think of the horses and cows that came from Central Europe; now they are all over the globe. They were transported by people.

Chicken and pigs were brought from Southeast Asia. So, in this way, people have assisted the spreading of animals.

Plants too, have been spread by people. Corn originated in Mexico but now is in the old world across the ocean. It is believed that corn would have died out if humans had not husked the corn and planted it.

Beans, tomatoes and potatoes originated in South America. But people in Europe depend on them today. Wheat, cherries and grapes were transported from the Tigris and Euphrates Valleys. Apples grew exclusively around the Caspian Sea. Think of where apples grow today – and all of the varieties of apples.

From these examples, we can see that humans, too, must be considered active agents of evolution. Look here! (center point of people circle)

Something has happened. As people have lived through the ages, taking from the environment and working on it, in order to fulfill their own physical and spiritual needs, they have created another structure on nature. It is a super-structure which Dr. Montessori called a *Sopra Natura*. Otherwise called a supra-nature.

Today, we have reached a point where people are very dependent on this super-structure

built on nature.

Think of the pencils we use in class. How could we work without the pencil. What about our clothes? Our food bought in the grocery store. Could we get along without these?

It is true that sometimes we talk about “getting back to nature.” But, think about how nice it is to have water delivered to your hands with a turn of the faucet. How wonderful it is to have light at the turn of a switch.

What about sending our messages across the oceans with a push of a button!

Even if we were able to give up those conveniences, what about having to give up all the beautiful music that has been created by different people? Or having to give up poetry or literature – those things that sustain our spirits?

So it is in this way that humanity depends on the supra-nature (point to center of people circle)

What is very exciting about it is that the supra-nature has been created by many many people over great lengths of time, some of whose names we don't even know.

For example, we don't know who invented the wheel. What about fire? Who discovered that? Some names we know but others we do not.

The other exciting thing about the supra-nature is that it does not belong to one person or one group but is the possession of all peoples.

That is why we have these arrows going back and forth – between all peoples and this silver supra-nature in the center of the people-circle.

Epilogue

This is not the kind of chart they want to make. But leave it out and let them reflect on it. It leads to much serious discussion among the children about which they are very serious.

February 23, 2005

Ecosystems

Biologists define an ecosystem as a system of organisms depending on each other and on the features of the non-living environment. In simpler terms an ecosystem is a community of living beings and the environment together which form an integrated unit.

Ecologists have identified **two factors making up an ecosystem. The first factor is the physical factor, also known abiotic factor – meaning without life.** The second factor is the non-physical factors – or biotic factors – living creatures. Make the children aware there are two kinds of factors that make up any ecosystem. The children have already experienced some of these in biology and geography. Let's look at abiotic factors.

One abiotic factor is water. The children have a lot of experience when it comes to the role of water. They know it is an important factor in geography, as well as in botany and zoology. They have seen this in the work of the river, how water lays down deposits, and that water is an agent of erosion. They also know that water profoundly influences living organisms.

In the **Needs of the Plant** experiment, the children notice that plants cannot even germinate without water. They see that water is a medium in which a variety of organisms live. They see this in an aquarium, sea-life or in pools of standing water that breed organisms. They know water has an influence on the climate – discovering this from the geography charts – such as the cycle of water. In demonstrations of plants, they know water is given off from the leaf. They know that water is returned to the environment by the animals through the deposits of waste.

Air is another abiotic factor. Air influences the environment and so it affects the ecosystem as well. It is a medium of existence for both land and water-surface organisms. The children know about **carbon and oxygen that are in the air.** They may even get involved in the carbon cycle and oxygen cycles. The children encounter these notions on the Leaf chart.

Then the **soil is another abiotic factor.** It is another factor affecting the organisms that live in it. How much moisture the soil holds is an important factor for the kinds of life it will support. When the children looked at the chart of vegetation, they became aware that more plants grow in moist soil than in dry soil.

The **last abiotic factor to consider is the sun.** The children know from their work with the role of the sun that it provides light and heat. From all of the impressionistic charts they have worked with, they know **the sun makes water evaporate, causes air to flow, creating climate patterns and keeps the earth warm** since the heat is trapped by layers of atmosphere.

So children have had a lot of experience with abiotic factors. So when formally speaking to them about these points, we merely remind them of what they've

experienced and know.

We also take up biotic factors.

Let them know they are all the living organisms that live in a certain community. Usually the factor that brings these organisms together is food. So, introduce to the children the food chain so they realize that it is one organism feeding upon another, which feeds upon another, and so on. When introducing the **food chain, one must speak about the producers and the producers are the green plants** – because they can produce their own food.

The **next link on the food chain are the consumers**. There are three orders of consumers: first is the **primary consumer and this is the animal that eats plants** – the herbivore; secondary consumer is one that feeds on the primary consumer and is the first carnivore. Then there is the tertiary consumer, which feeds on the secondary consumer and is the second carnivore. Also, a part of the food chain are the **decomposers**, which are all of the bacteria that feed upon decomposing matter.

In speaking about ecosystems, one must speak of relationships. In biology, that relationship is called symbiosis. Symbiosis comes from two Greek words, '**sym**' meaning **same**, and '**biosis**', meaning life. It is the relationship of two or more different organisms in close association with each other; they may or may not be of benefit to each other.

There are **different kinds of symbiosis**. One kind is called **commensalism**. This comes from two Latin words '**cum**' meaning **with** and **mensa**, which means **table**. They share the same table. This is where **one organism benefits from the relationship but the other organism is not effected**. Examples include the Remora fish, which hangs onto a shark, catching what the shark misses as it eats.

Mutualism is another kind of symbiosis. **Both organisms benefit** from this relationship. The Barber shrimp has an antennae and throws out a signal to the fish and they queue up to have their bodies pruned and cleaned. The shrimp gets a good meal and the fish are physically maintained. Lichens is another example – made up of a fungus and an algae – that mutually benefit one another.

The last type of symbiosis is **parasitism**. In parasitism, **one organism benefits and the other is harmed**. Each human body is an ecosystem with all kinds of symbiotic relationships – mites in the eyelashes, for example.

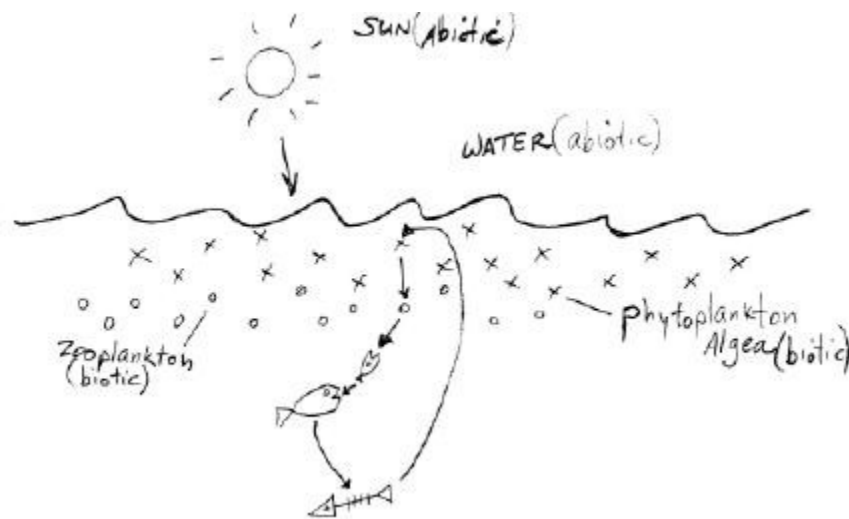
When talking about ecosystems, let the children know there can be **water ecosystems and land ecosystems** – and introduce just one the the children, any one. Let them discover the others on their own. If a water system is selected, one must choose between a fresh water and salt water system. If selecting fresh water system, again select between a lake and a river. If choosing a lake, one will again choose between life on the shore and life in the middle of the lake. If choosing salt-water systems, select an inter-tidal zone, around the shore or else the life that lives in the middle of the ocean.

When introducing an ecosystem to the children, it can be done **using a simple drawing** on a board or piece of paper. Keep the illustration simple. Draw abiotic factors first and then add in the biotic factors – followed by arrows (going in the correct

direction) that show relationships. And make sure the arrows ultimately go in a large circle that shows the larger cycle of life.

Sample Story

Here we have the sun. It provides for the phytoplankton (bits of different kinds of algae), the light it needs to make food. (draw arrow) The zooplankton feed upon the phytoplankton. (draw arrow) The tiny fish feed upon the zooplankton to survive and grow and so provide for the larger fish. (draw arrow) One day the larger fish dies and the decomposers feed upon the corpse of the fish and break it down into nitrogen, oxygen and carbon – the larger fish, when it dies, provides for the decomposers. The nitrogen, oxygen and carbon provide the plants the minerals they need. (draw arrow).



Adolescent Education

Have a meeting with parents and staff to announce sex education will be covered. They may want to decline to participate. Send a note home as well prior to the lesson. The lesson depends on the moral climate of parents and faculty – what is acceptable.

The early lessons focus on anatomy. The children are interested in the changes in their bodies more than how babies are made. They want to know they are normal and that everyone changes at a different rate and has different results.

You can separate boys and girls if needs be but it is preferable to keep them together when possible. Have a container the children can put anonymous questions. They may not ask questions otherwise. It is surprising how many children and/or parents

are uncomfortable about speaking about sex education.

It is important today given problems with AIDS. They need facts from knowledgeable adults. Let the children know of their right to not be touched. Have a library of books available for help; "Period", for example. "What's Happening to Me?", is another. Birth control can be approached in the context of sexually transmitted disease.

October 20, 2004

Botany Nomenclature Material

We've spoken of nomenclature material with all previous lessons. They are used with children who are interested in reading or interested in botany. We should not insist that every child use it.

If we insist all children use these materials, botany dies because it takes the fun and excitement out of it.

The material has definitions of the characteristics. What we do not want to do, is to have them copy the definitions out of the books. We encourage them to do all kinds of writing with botany. But their definitions should come from their own experiences. They make their own sentences, their own words and generally, they are not very scientific – but they make sense to them. That is a beautiful piece of work – when they own it.

When you present this material, present it with a real specimen. The new children in the elementary class will recognize this material since some of it has been in the primary class. But in the elementary classroom, it has been expanded so there is much much more to discover.

Description

There are three different sets (or racks). The characteristics in each of the racks are organized from front to back going from the whole - then working toward the back of the parts. This is the case for each rack.

The characteristics that are used for scientific classification are placed at the front of the rack. Within each rack, one finds the different parts of the plant color-coded. For example, the leaves are on green book ends, the flowers are red, the stems are on a light brown. Each part has its own colors. It's not necessary to use the same colors, however, it makes sense to the children to have leaves on green and flowers on red.

Within each rack, you have a coding. The first rack for the leaf is coded 'L1'. The back of the cards are all labeled and the tickets are all labeled as well. Do not overlook the labeling on the tickets for your own sanity. The codes are to help the children put the material back in order.

Before we leave the institute, get the master pictures (black lines only). We need to color at least one card so we have it for future reference.

Rack One

This has a set of pictures. The first has the name attached to the picture. The second set has the same picture but has no name. To go with the unnamed picture, there is a ticket written in black ink – with the name on it. This rack is used for children who have never had the material in the primary and for children who don't read well.

Second Rack

It has definitions. It has the name or some other aspect missing. On tickets, written in red ink, you have the name or the missing aspect. To correspond with that definition, you have a booklet. The booklet will have a picture on the left side and the characteristic will be isolated in color. (You have the leaves, the blades, the petiole, etc.) On the right side, you have the complete definition and the characteristic or name is isolated in red ink.

Third Rack

This one should only have a few characteristics, a few definitions. It consists of tickets on which the definitions are put. They are cut up definitions but are put on tickets of the same size. The name or aspect is on the same size of ticket but is written in red ink.

What we must never do is take a definition that is cut up and use it like puzzle pieces because the children don't think; they treat it only as a puzzle.

Procedure

How to use the first rack material

The material to use first is the set with the name. But you should also have a real specimen. You introduce the children to each of the cards and each of the terms, pointing them out on the real specimen at the same time.

Do a **three period lesson** with the children but because they are second plane children, you want to **make the second period game-like and active**. Get them running outside to collect leaves, to look at veins and so on. When the children have had experience with the first set of cards, move on to the second set.

The second set has no name. They can place the ticket with the name to the corresponding picture. They can control their work with the first set of cards that has the name written on it.

How to use the second rack material

This comes after they have experience with the first rack plus their own exploration. They can take the **tickets and make them into heading cards on the floor**. Then they can take each definition, read it and place it under the correct heading card. They can control their work with the booklet.

How to use the third rack material

The children already have experience with the characteristics and the parts. They enjoy botany. Take the name tickets and lay them out as headings. It doesn't matter the order in which they are laid. "We are doing the plant – close your eyes!" Hand out the tickets and let them do each one.

October 20, 2004

Botany Command Cards

The age is probably 6 ½ to 7 for this activity since they must be responsible to a certain degree. The over-arching purpose for these cards is to do independent work, to structure their work and build responsibility and follow-up skills.

The children have had lots of experiences with you in botany. But with the command cards, they have an opportunity to do independent work and it allows them to repeat some of the work you've done with them. It also allows them to dwell upon the ideas that have been presented.

The botany commands provide a language activity for the children. So, when they do a command activity, there has to be writing associated with it. They can set their command up, observe it and then they can write the conclusion to what they saw. When they write the conclusions themselves, they will need to form sentences, punctuate carefully and spell correctly. In this sense, there is a connection between botany and the language study.

These activities they do with botany commands **can be looked upon as practical life activities**. They have to be exact in how they do the activity. They have to follow a sequence and it is meaningful work for them.

The procedure for doing the botany commands is essentially the same as it is for the geography commands except it is less rigid. There are no dangerous chemicals involved in the botany commands.

Just like the nomenclature material, the botany commands should not be over-emphasized in importance. What is important is the exploration and the discoveries that the children make on their own.

Description

The **first set of commands** are cards that have written on them the **subject, the material needed and the command** – how to do the activity. This is the command.

The **second set** is made up of the **statement cards**. The statement cards have the conclusion of what happens – or the **results** of the command.

The **third set** of cards have both the **information** on the **command card** as well as the information on the **statement card**. This third set is used as the means for **control of error**.

Procedure

Initially, both teacher and children are involved in the activity. The teacher does most of the activity and the children are principally involved in the setup and the cleanup. The cleanup is critical and we need to expect the children to do it always.

The next time you take out the material, the children are in charge of the activity. You just keep an eye on it – especially the cleanup. The third time they use the material, they can do it without the teacher at all.

A child can **select any card he wants** and he selects it from the command cards. He **sets the activity up according to the command** and then he waits to see what happens. Sometimes the results occur in a matter of hours and other times it is a matter of days. So they have to keep an eye on the activity.

Once they get a result, they should go to the second set of cards, sort through the statement cards reading each, until they **find a statement that matches their results**.

After they've picked their statement, they can go to the third – commands and statement – and **find the command card for the experiment they chose**, to be sure they have chosen the correct statement.

October 20, 2004

Use of the Microscope

If you are lucky you will have access to a microscope in the classroom. Not all can afford one. There are a number of ways to get the use of one however.

Sometimes there are parents who have microscopes in their homes and they are happy to lend them to you – just get the word out you need one. In some areas, there are children's museums that will also let you check them out to borrow for a time. And then, you can buy your own.

If you buy a microscope, do not buy one from Toys R Us. It's a waste of money. There are places that refurbish microscopes and sell them used. Sometimes you can get used microscopes from hospitals. You can get used microscopes from universities. So there are places to find them at a discount.

If you can buy a new microscope, get a Nasco catalog or Carolina Biological is another place to shop. Nasco is generally better priced.

If you don't have money to buy one, the school can fund-raise. Parents will support this kind of fund-raising. Then you have money so you must choose what kind to get.

You need three or four hundred dollars to get a decent microscope. There are **two kinds: dissecting microscope and compound microscope**. If you have to choose one or the other, choose a dissecting microscope. It can be used for gross things – large things like insects, leaves and crystals. It will become more useful. Compound microscope is more useful for looking at minute things such as an amoeba or a blood cell.

If you are in a 6 to 9 year old class, a dissecting microscope is very useful. The compound microscope is more suited to the 9 to 12 years old class.

If you are in a large school, you can usually share a microscope. However, it is nice to have your own because you can see that it is well taken care of. You can insist it is taken care of.

The microscope is a scientific piece of material and requires attention. People should never touch the lenses. When you touch the lenses, you leave oil on the surface. Whenever you need to clean the lenses, use lens paper – not tissue paper nor cloth. There is a special lens paper that is superior.

When the microscope is not in use, it should be covered at all times. It is a habit that you must insist the children develop. This is where follow-through on your part is important. You have to notice that it is uncovered and track down who left it so, to make that person aware that it cannot be left this way. If that person cannot raise their awareness, then take the privilege of its use away from them.

Another thing children are tempted to do is to leave their specimens on the microscope. And because children often work in groups, one might think others have taken care of things when in fact, no one has taken care. Track them down and if you are consistent, it becomes a matter of course for the children – to clean up consistently. This applies to everything – not just the microscope.

Presentation of the Microscope

Because you want children to appreciate this expensive piece of material, you want to give them a lesson on how to use it. Tell them about the dust cover. Tell them the microscope must be covered at all times when not in use; dust is the enemy to the microscope.

Introduce them to the parts of the microscope. The place where you put your specimen is called the **stage**. You can fix the specimen on the slide with these clips. The pieces you look through are called the **oculars**. There bi-ocular and mono-ocular. The parts that magnifies are called the **objectives**. In some microscopes, they come in different strengths. Let the children experience what it feels like to move the objectives until they lock into place.

Show the children how to focus in on the specimen. You usually have a **coarse adjustment and a fine adjustment**. When you focus in on your specimen, use the lowest power objective in place. With the coarse adjustment, focus-in on the specimen. Once you get it in focus, then you can move it to the next highest power, continuing to focus in with the coarse adjustment. Then once in focus, you can use the fine adjustment so the image becomes precise.

On some microscopes, there is something called **oil immersion** associated with the highest power objective. This is a special objective not found on other microscopes. You won't need that unless you are looking at cells, like red blood cells. You would put a small bit of oil onto the slide and then put that particular objective into place, and then make an adjustment with the fine adjustment controller.

You can clean the stage with rubbing alcohol. Usually one uses a petrii dish so rarely do you need to clean the stage.